

*One for the buzzard,
One for the crow,
One to rot, and
One to grow!*

- Fay Yauger

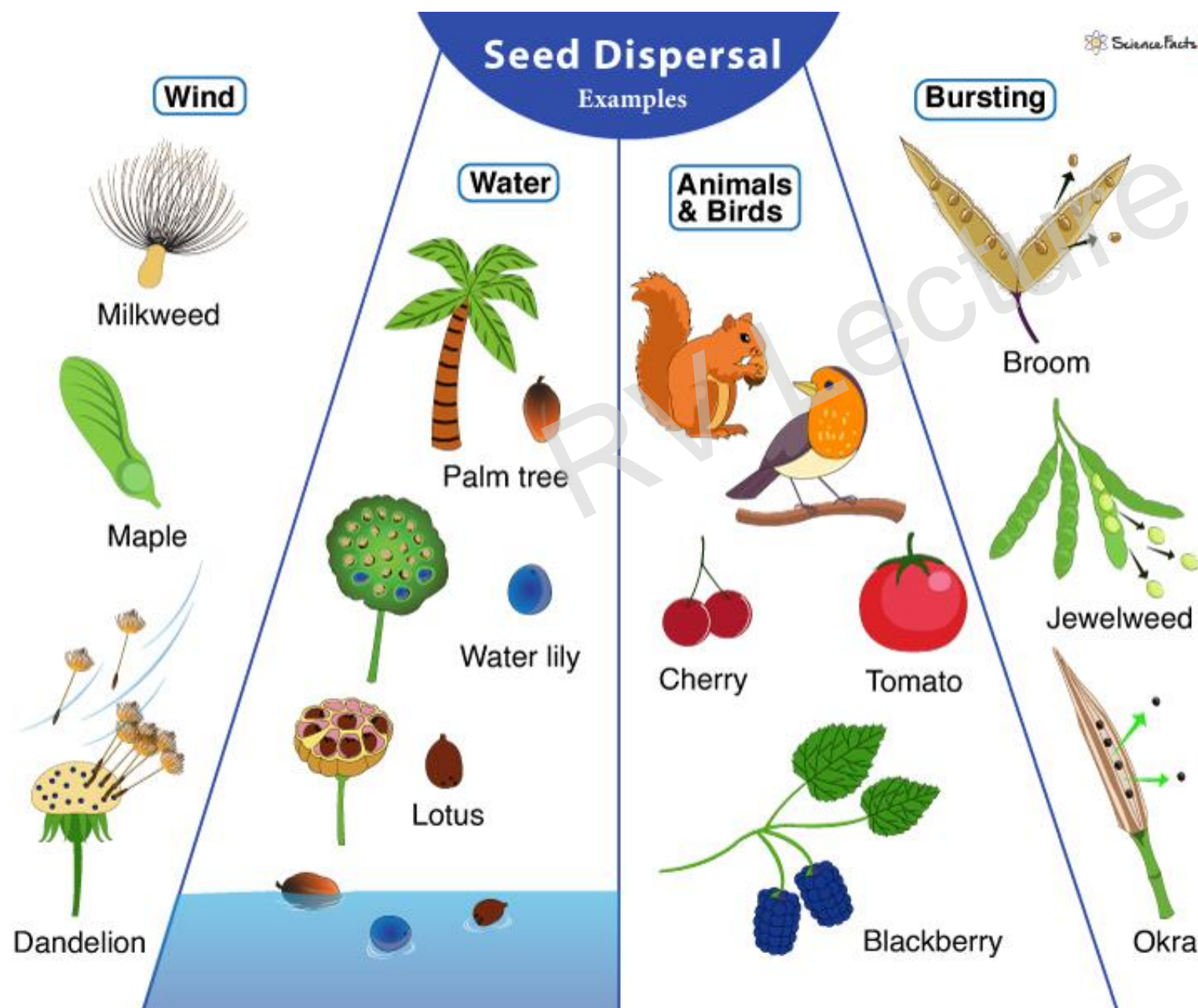


The seed itself is designed to disperse and scatter, using other forces of nature-wind, water, insects, birds, and other animals.

Seeds have been, and still are, the mainstay of the world's diet. The Poaceae, or large-seeded grasses, collectively known as cereals, contribute more food seeds than any other plant family. Cereal grains comprise approximately 90% of all cultivated seeds. They provide people with their most important source of carbohydrates, as well as some protein and other vital substances. As in ancient times, rice, wheat, and corn are the three major grains. Oat, barley, sorghum, millet, and rye are other important food and feed grains.

The second most critical food family, the Fabaceae, provides crops such as peanut, soybean, lentil, bean, pea, and chickpea.

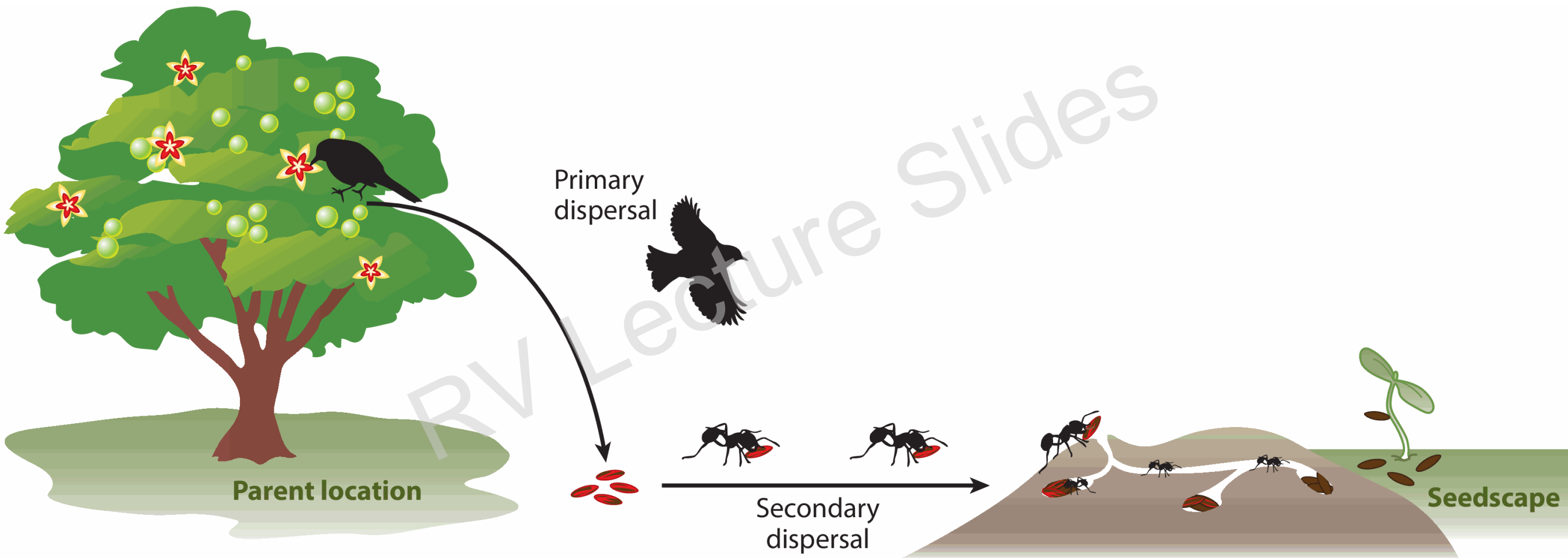
In addition to being used directly as foods, seeds play other roles in human diets. Many seeds are used whole or ground as spices. The popular beverages coffee, cola, and cocoa are derived from seeds. Beers and ales are brewed from barley, and whiskey and gin are distilled from fermented mashes of cereal grains. Edible oils are obtained from seeds of corn, soybean, canola/rapeseed, cotton, peanut, coconut, palm, sunflower, and safflower. And seeds are used in the manufacture of some drugs and medicines.



How a plant grows from a seed



Seed dispersal, or the movement of diaspores away from the parent location, is a multiscale, multipartner process that depends on the interaction of plant life history with vector movement and the environment. Seed dispersal underpins many important plant ecological and evolutionary processes such as gene flow, population dynamics, range expansion, and diversity.



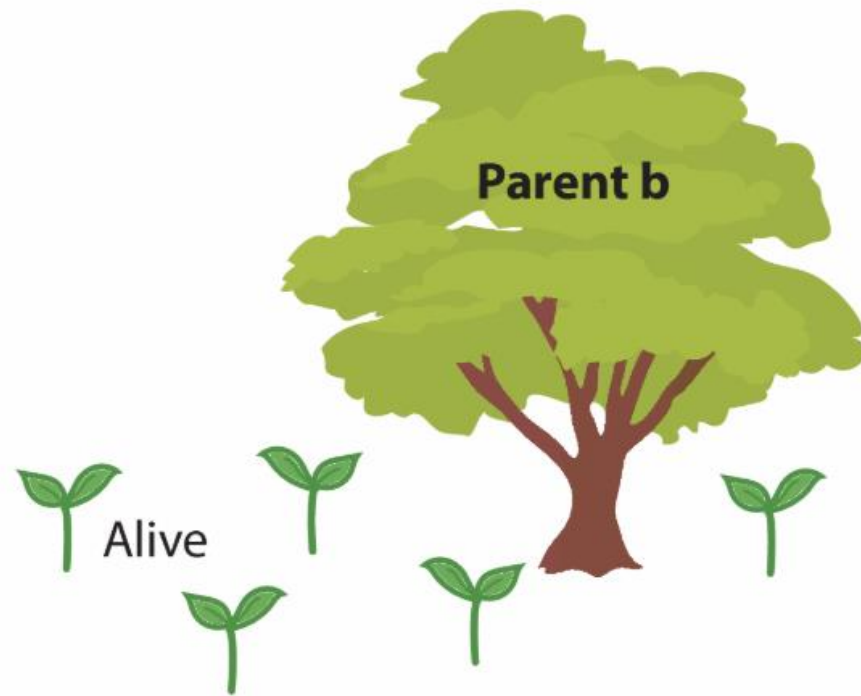
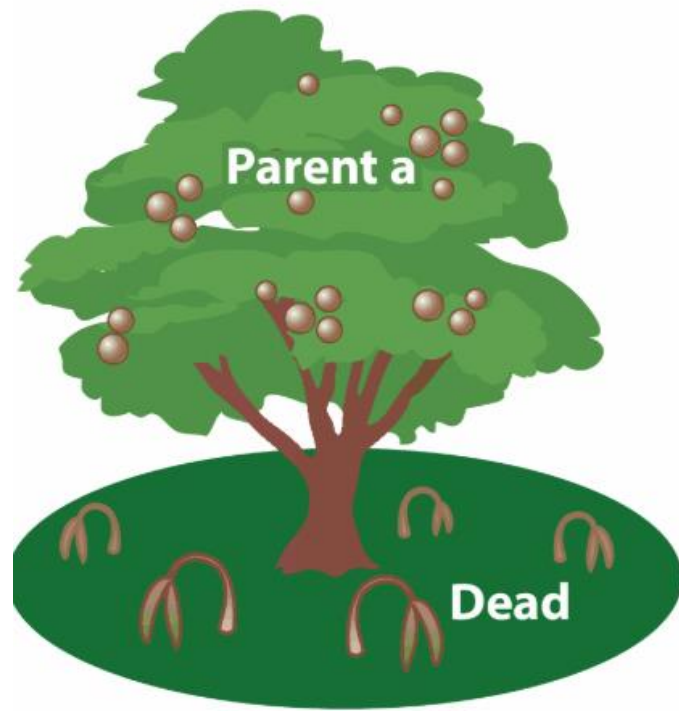
WHY DISPERSE? ADAPTIVE ADVANTAGES OF SEED DISPERSAL

Dispersal of organisms is costly in energy, time, risk, and opportunity. Then why do plants disperse?

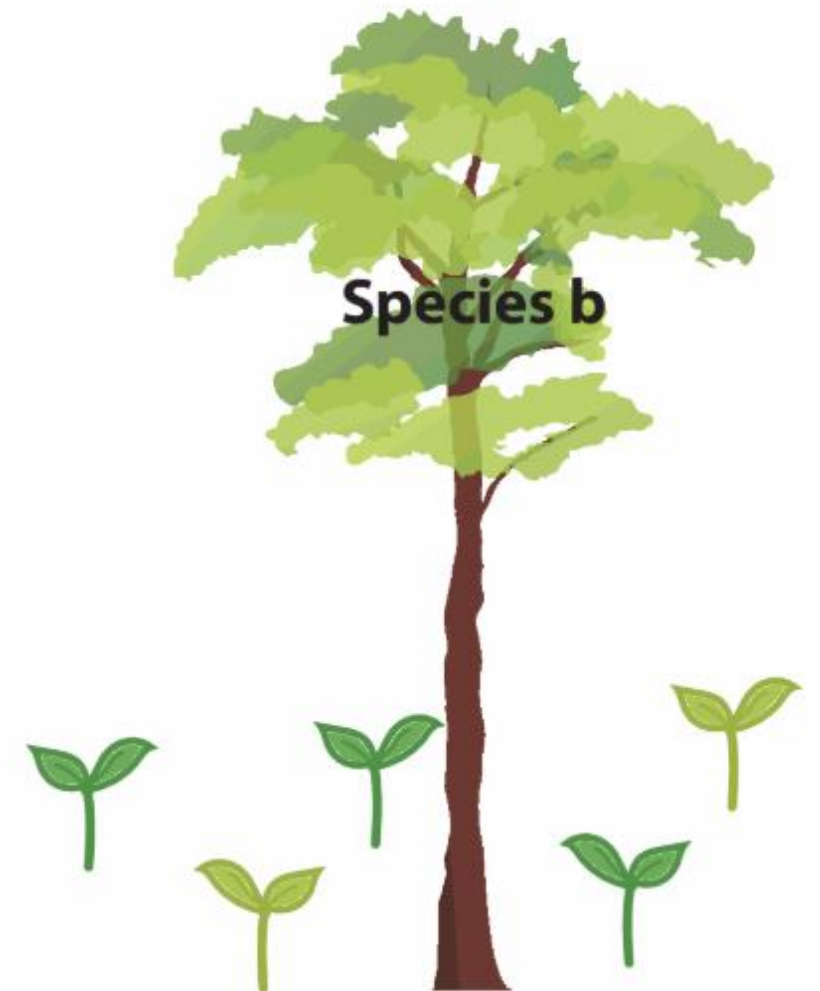
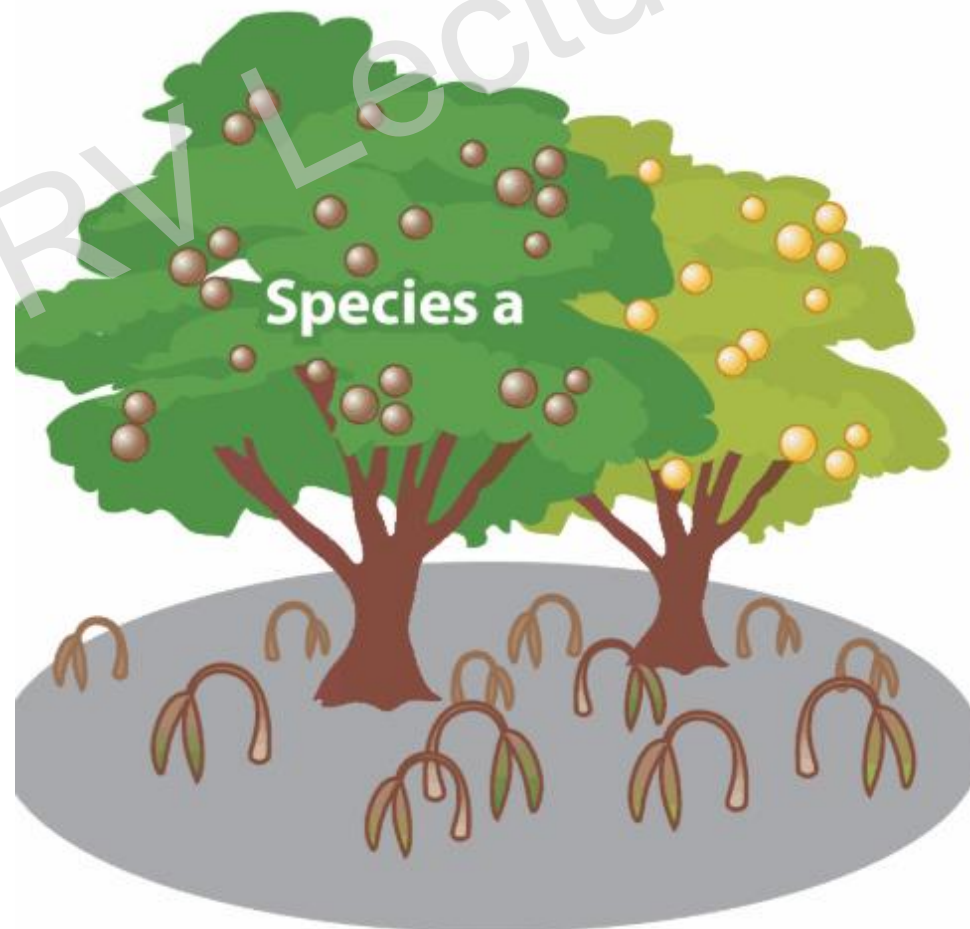
Evolutionary theory postulates that the loss or death of propagules during dispersal episodes (cost of dispersal) should select against dispersal.

The major selection pressures that increase dispersal, typically measured in theoretical studies as the proportion of dispersing offspring or the distribution of dispersal distances, are generally related to competition and spatiotemporal variation in local environmental conditions

i Sibling competition or parent–offspring competition



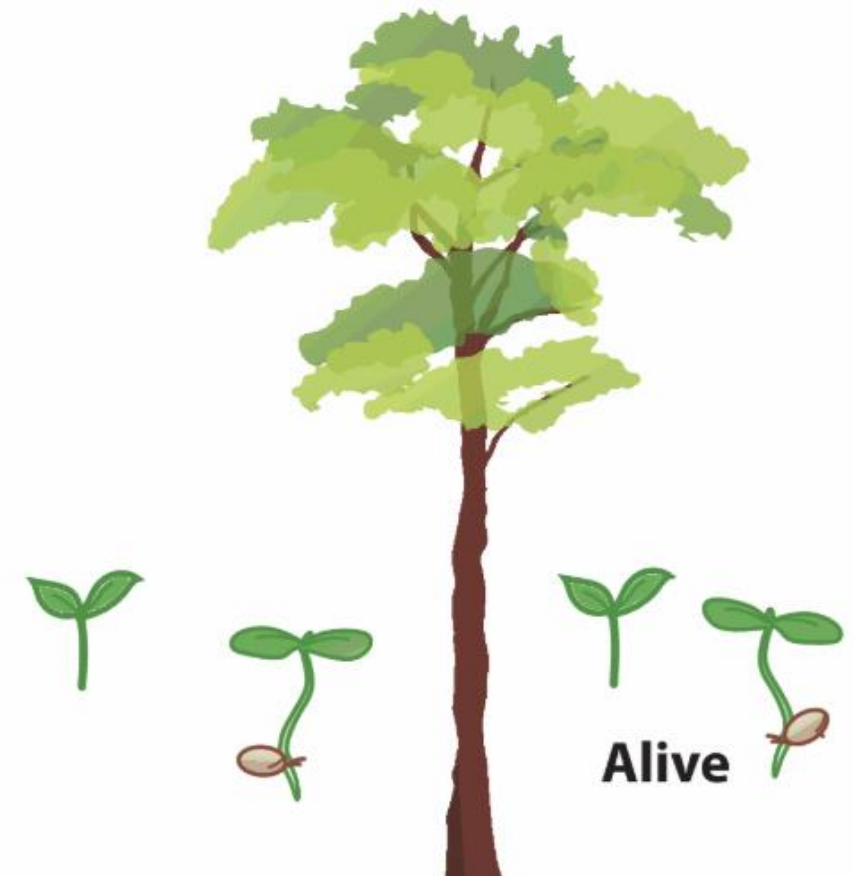
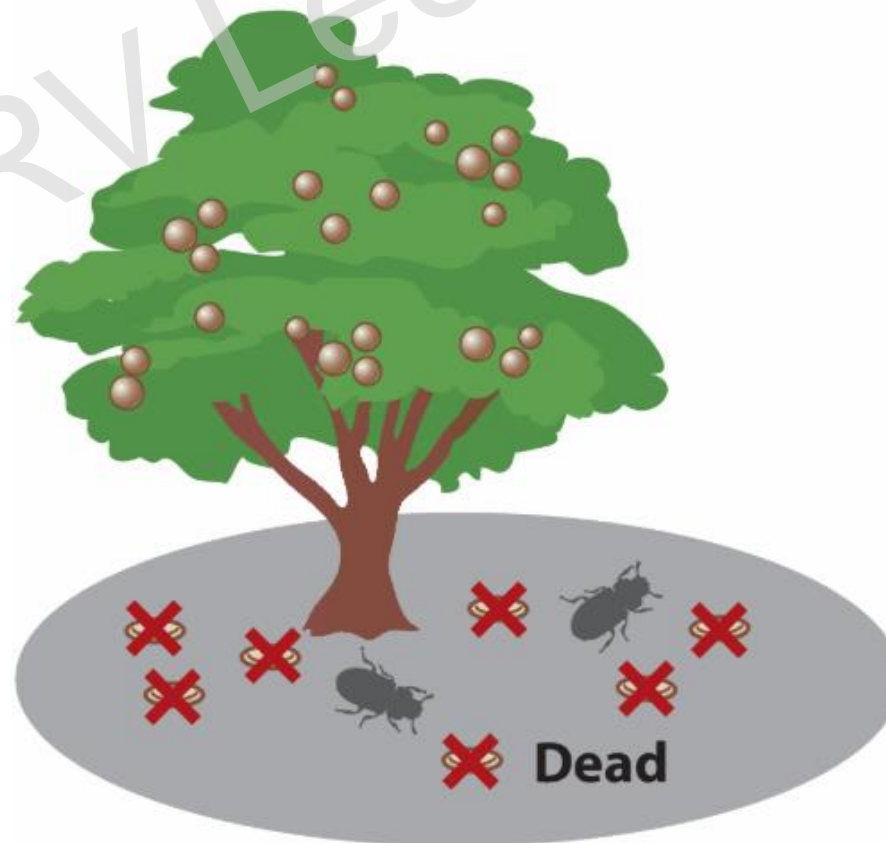
ii Conspecific competition



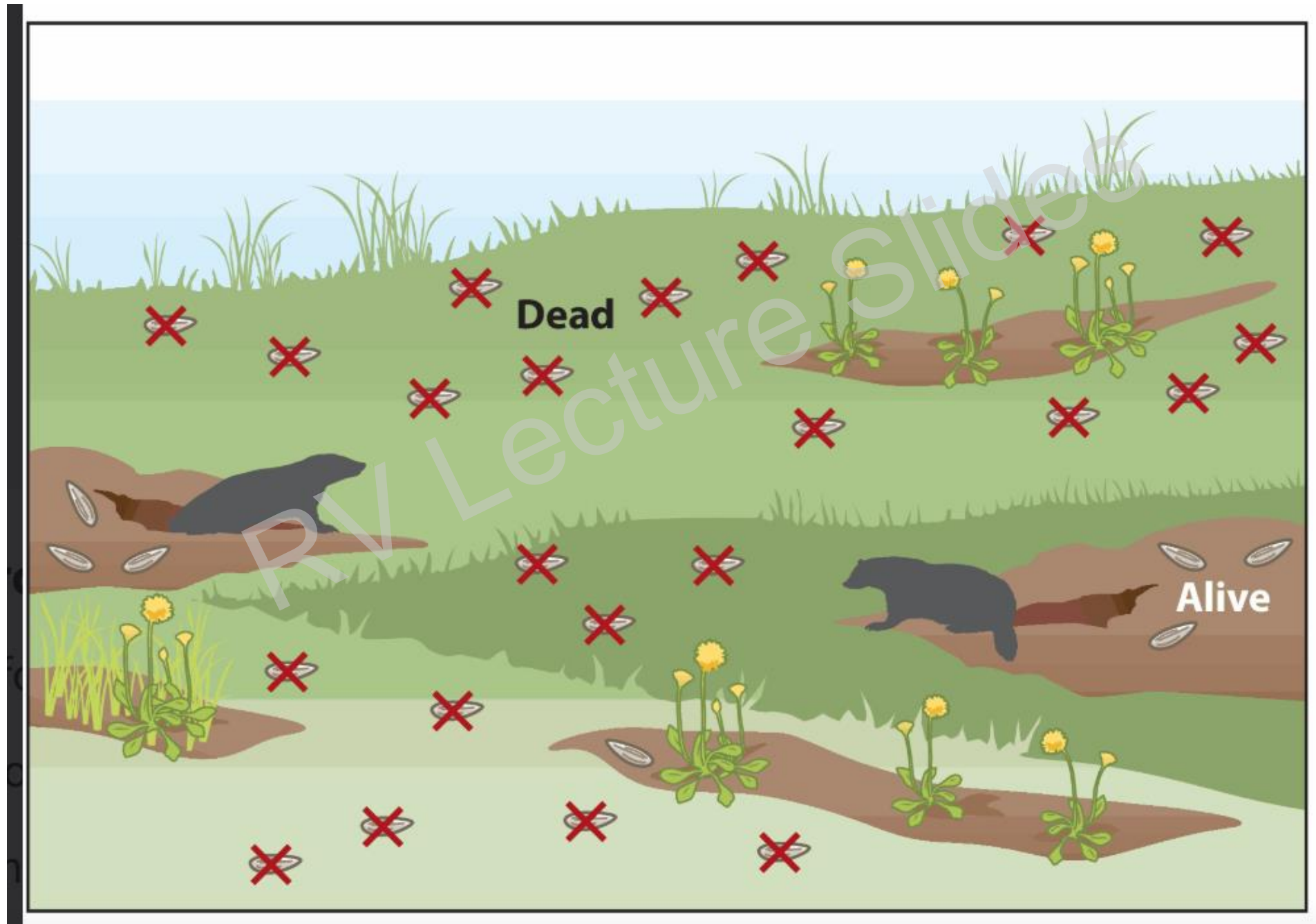
i Soilborne pathogens specialized on maternal plants



ii Insect seed predators specialized on host plant species



Individuals with higher dispersal can have a selective advantage that allows them to take advantage of ephemeral habitats, such as treefall gaps in forests, polluted landscapes, or disturbance mounds created by animals

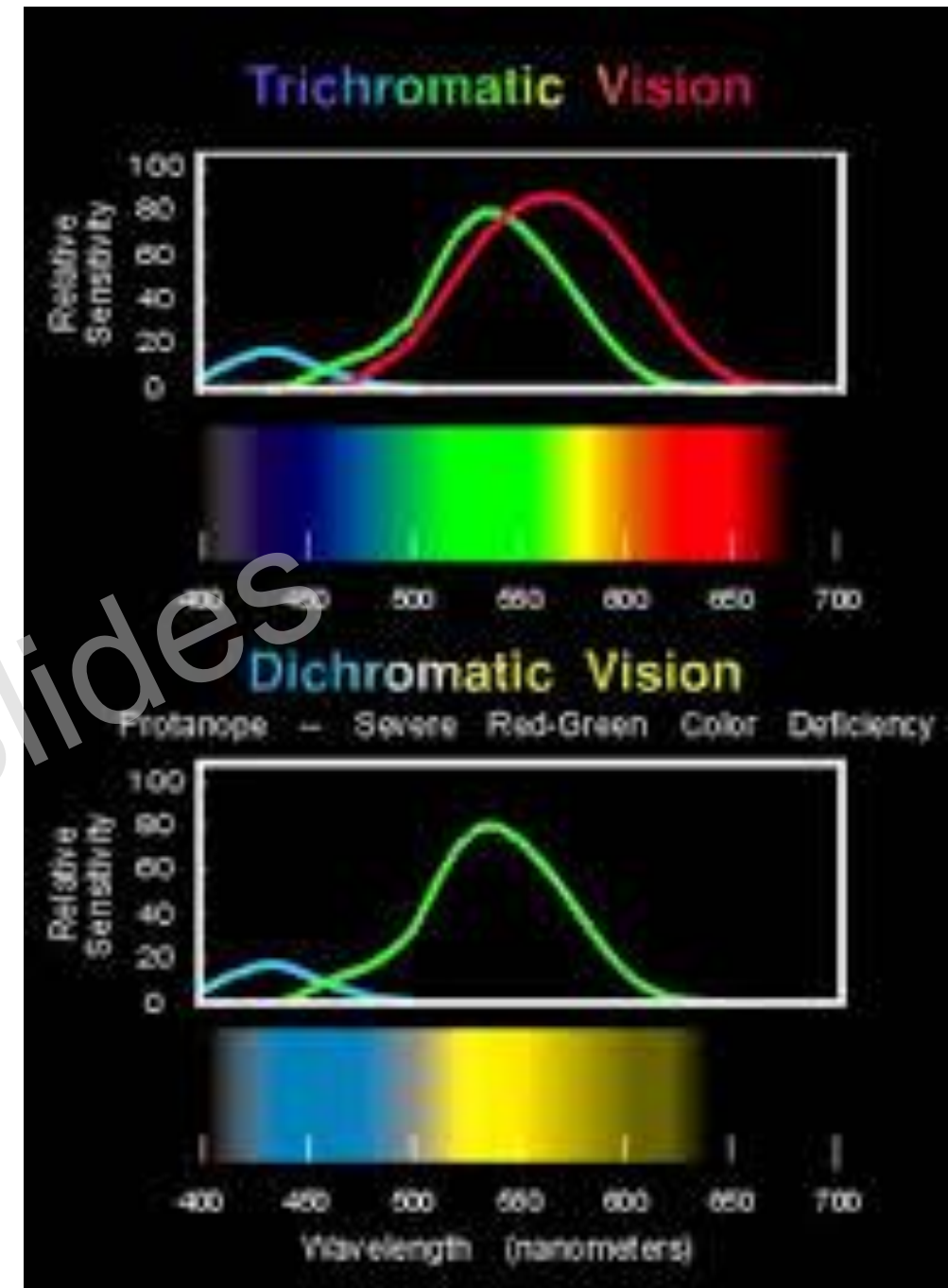


Many flowering plants rely on **animals to disperse their seeds**, and ripe fruits often advertise themselves using **visual cues**—especially **color**. For mammals that eat fruit, vision can directly shape what they find, eat, and disperse.

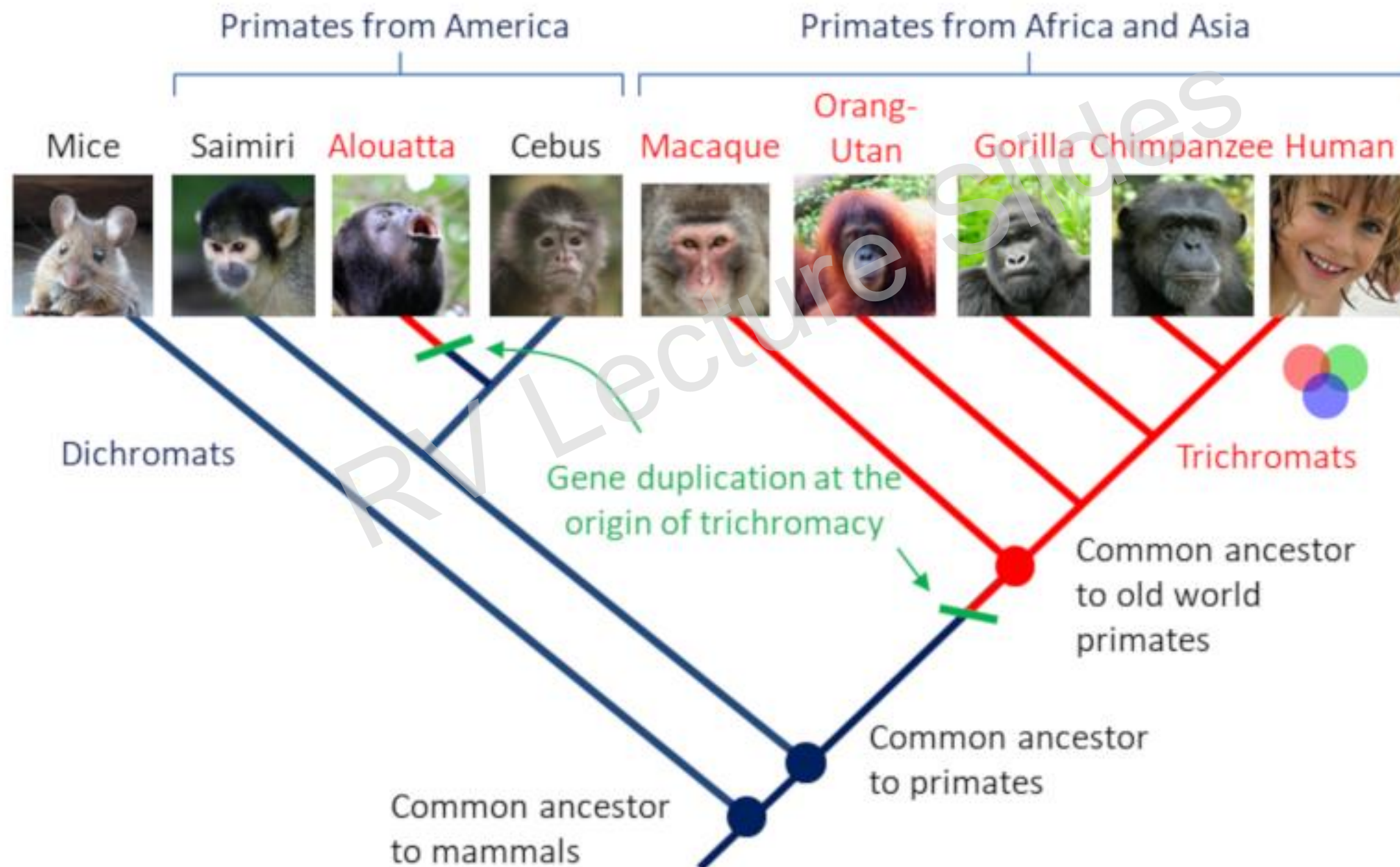


Nocturnal Mammalian Ancestors: During the age of dinosaurs, early mammalian ancestors were primarily nocturnal to avoid predators. Color vision is less useful at night, so natural selection pressure led to the loss of two of their four cone opsin genes, resulting in two-color (dichromatic) vision. Most modern mammals, such as dogs and cats, remain dichromatic.

Primate Re-evolution: Millions of years later, some primate ancestors became active during the daytime again. A key evolutionary event occurred in a common ancestor of Old World monkeys and apes: a duplication mutation of the existing green-sensitive opsin gene, followed by further mutations that shifted the new gene's sensitivity to longer wavelengths (red).



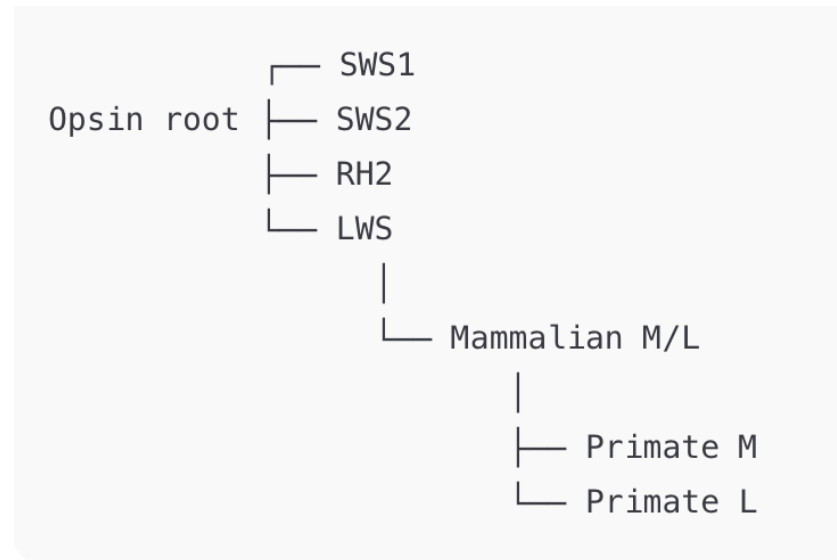
Ancestral vertebrates possessed multiple cone opsins, but early mammals lost long-wavelength sensitivity during a nocturnal phase; primate trichromacy represents a derived adaptation that functionally resembles the ancestral condition rather than a true evolutionary reversal.



Early vertebrates had **multiple cone opsins**:

- **SWS1** → short wavelength (UV/blue)
- **SWS2** → blue
- **RH2** → green
- **LWS** → long wavelength (red)

This allowed **broad color discrimination**.



In Old World primates:

- The remaining **M/L opsin gene** on the X chromosome
- **Accidentally duplicated**

Just a few amino acid substitutions (notably at positions like 180, 277, 285) caused:

One opsin to absorb slightly **shorter wavelengths** → **M (green) opsin**

The other to absorb slightly **longer wavelengths** → **L (red) opsin**

Early mammals went nocturnal.

Consequences:

- Color discrimination less useful
- Some opsins were **lost or inactivated**

Typical early mammal:

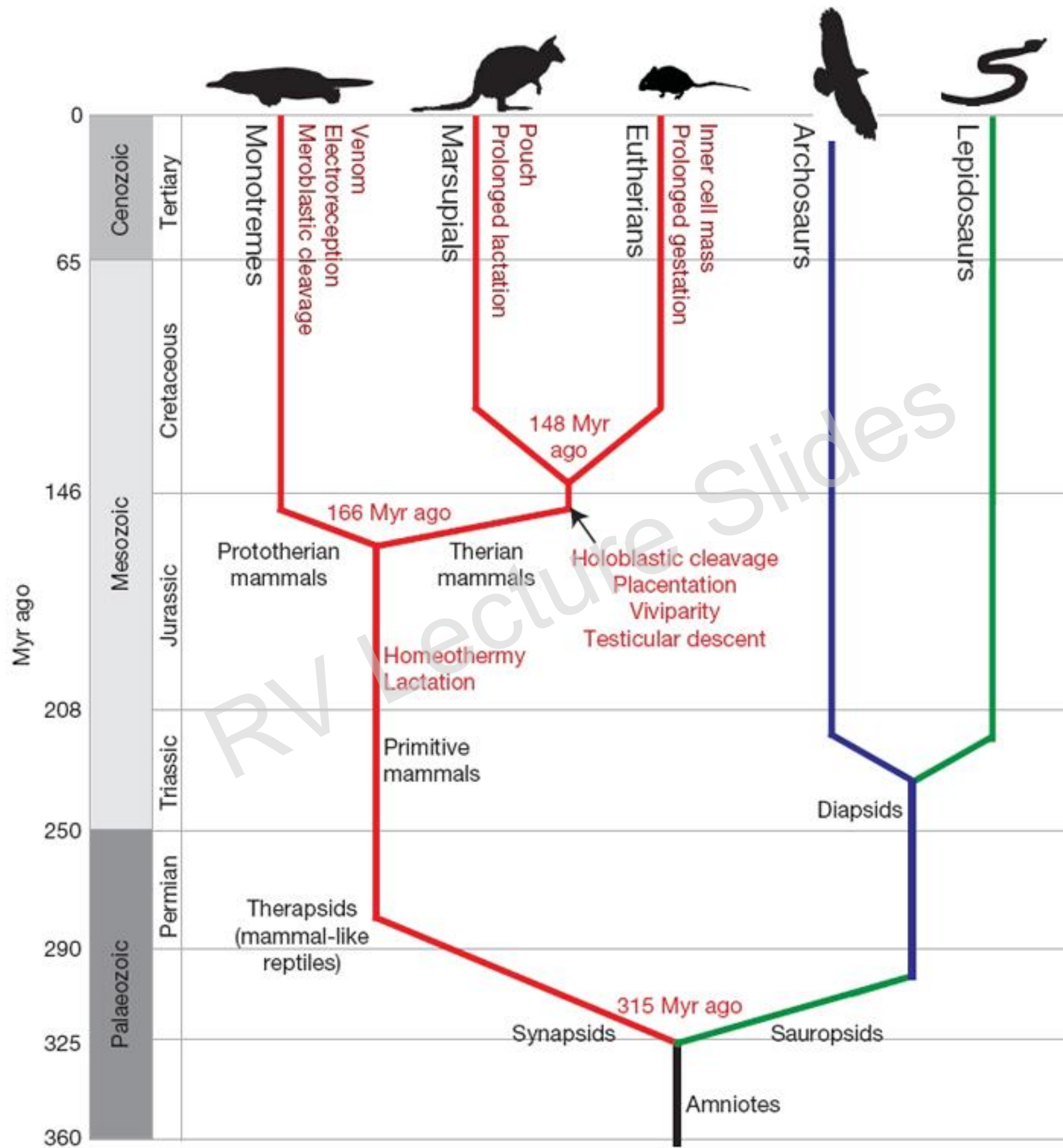
- **SWS1** (blue)
- **One M/L opsin** (greenish)

→ **Dichromatic vision**

Now primates had:

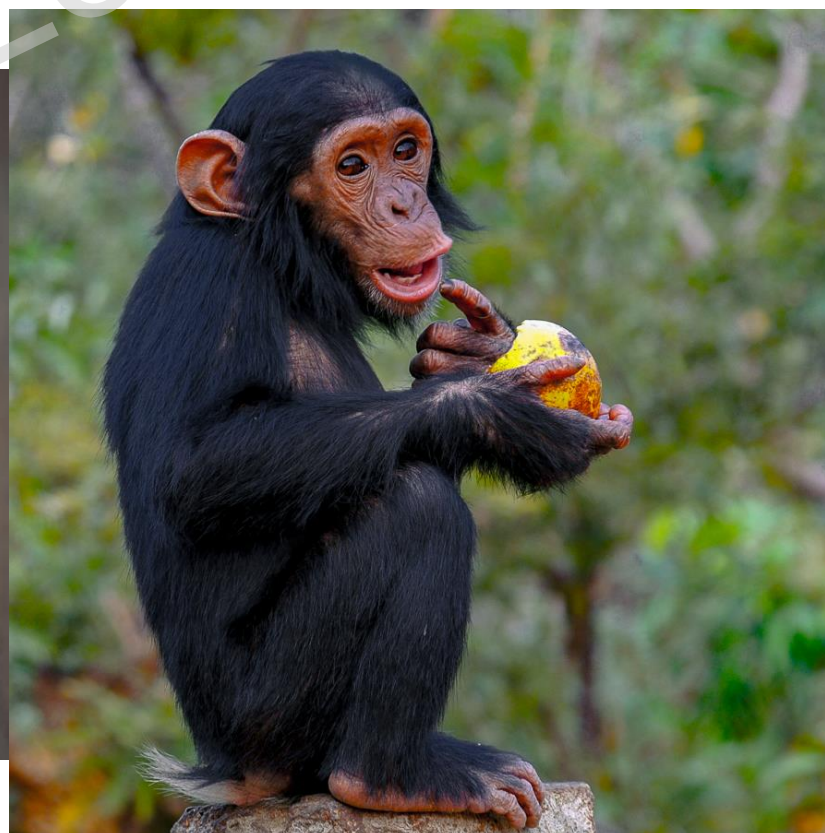
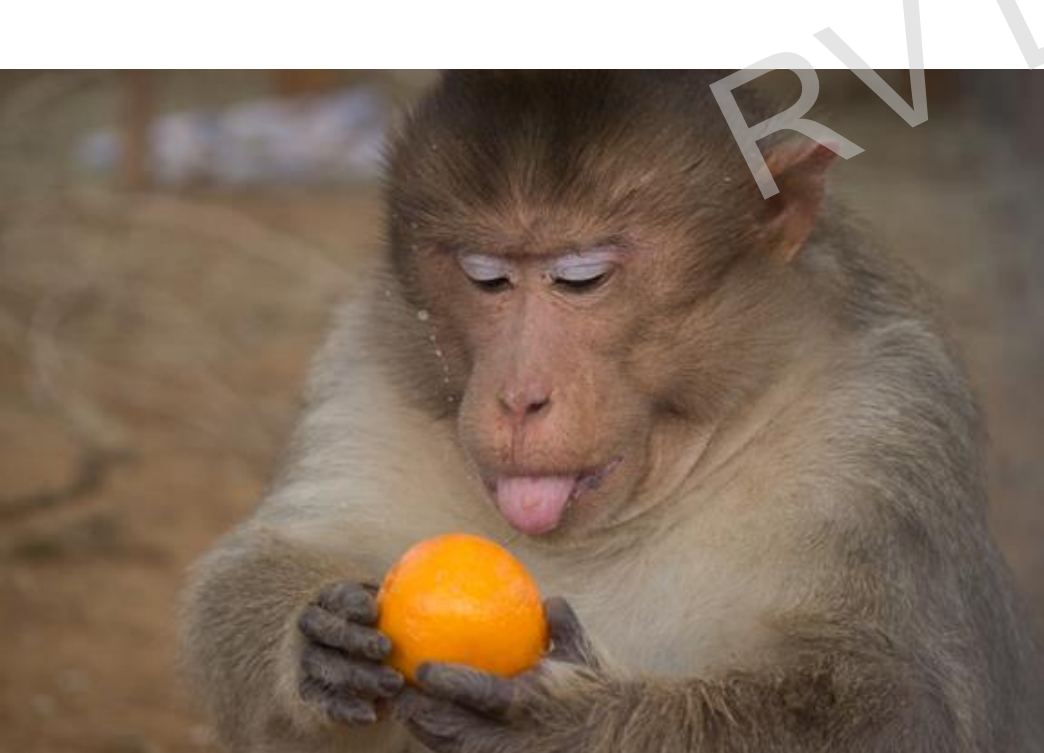
- **S cone** → blue
- **M cone** → green
- **L cone** → red

→ **Trichromatic vision**



Lineage	Color vision history
Early vertebrates	Tetrachromatic
Birds	Retained tetrachromacy
Most mammals	Reduced to dichromacy
Old World primates	Trichromacy via duplication

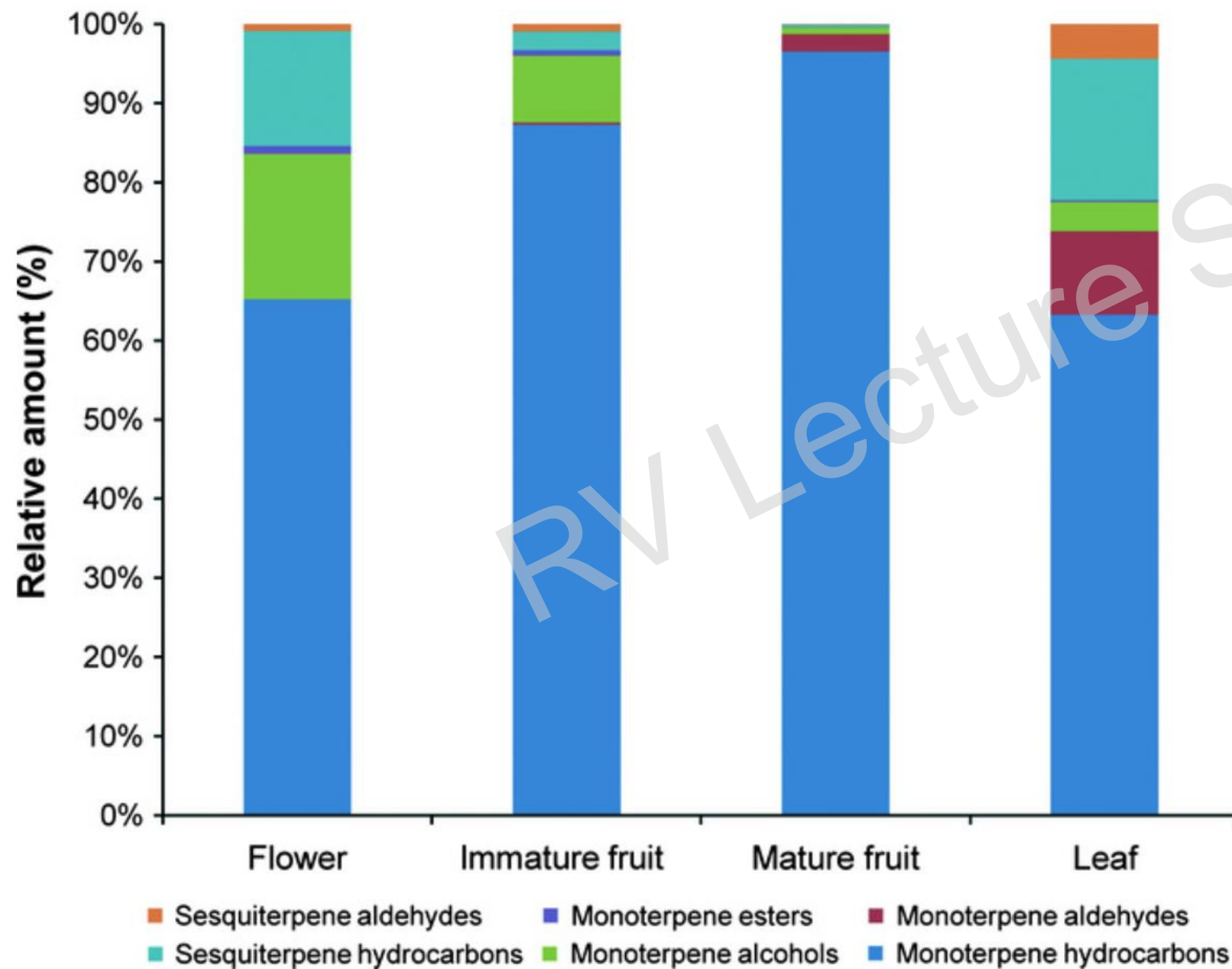
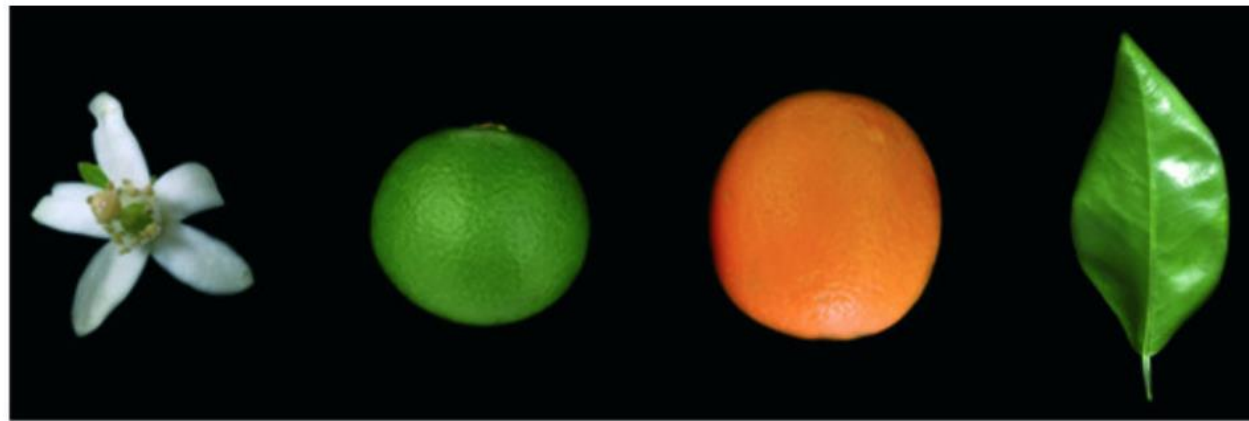
Disperser type	Dominant sensory cue	Typical fruit traits	Representative fruits
Birds (Ornithochory)	Bright color (UV–visible), shape	Red/blue/black fruits, small–medium size, low odor, exposed	Cherry, Blueberry, Mulberry, Fig (many species), Holly
Fruit bats (Chiropterochory)	Strong odor (night-active)	Dull green/yellow, soft pulp, hanging or exposed fruits	Fig, Guava (wild types), Banana (wild, seeded), Breadfruit
Primates	Odor + handling + moderate color	Medium–large fruits, fleshy arils, swallow/spit seeds	Jackfruit, Mango, Papaya, Sapodilla
Ungulates (deer, cattle)	Odor + bulk feeding	Fibrous pulp, tough seeds, gut-resistant	Fig (ground-fallen), wild berries, many palms
Rodents (scatter hoarders)	Odor + handling	Hard seeds or nuts, often buried	Acorn, Chestnut, Walnut



Can plants manipulate frugivores?

Mango peel and pulp have large amounts of volatiles, compounds known as terpenes that are attractive to many animals. They also have high sugar pulp that is a reward for the disperser.





In general, flowers and fruits release the widest variety of VOCs, with emission rates peaking before pollination and at ripening, respectively.

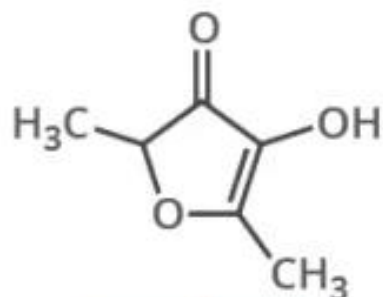
Flowers, leaves and fruits often show different VOC profiles, suggesting that their functions in different tissues or organs may also be different

THE CHEMISTRY OF MANGOES

MANGO FLAVOUR & AROMA COMPOUNDS

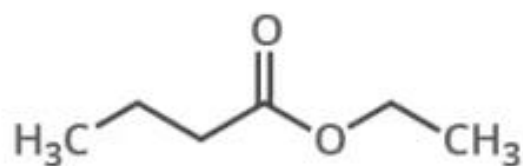
A large number of compounds contribute to the flavour and the aroma of mangoes. The cultivar, maturity, and geographical origin of the mango all influence the compounds present.

270+ VOLATILE COMPOUNDS
DETECTED IN MANGOES

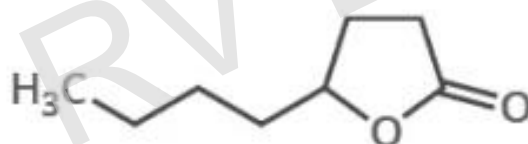


HDMF

γ -OCTALACTONE



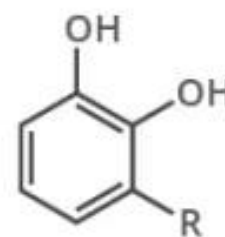
ETHYL BUTANOATE



Esters such as ethyl butanoate account for fruity notes in mango aroma. A major contributor to sweet notes is HDMF (4-hydroxy-2,5-dimethyl-3(2H)-furanone). Lactones such as γ -octalactone can lend a coconut-like aroma, while terpenes are also found in significant quantities and make minor contributions.



MANGOES & CONTACT DERMATITIS



Possible R groups

$(CH_2)_{14}CH_3$
 $(CH_2)_7CH=CH(CH_2)_5CH_3$
 $(CH_2)_7CH=CHCH_2CH=CH(CH_2)_2CH_3$
and others...

URUSHIOL

Mangoes belong to the same family of plants as poison ivy. Urushiol, a mix of similar organic compounds which are found in poison ivy and can cause a rash to develop on contact with the skin, can also be found in mango skin. This means that some people who are sensitive to urushiol get contact dermatitis when chopping or eating mangoes.



Sensory Bias: Animals exhibit inherent biases in their perception and preference for certain stimuli (e.g., specific colors or scents), which plants exploit to increase the likelihood of their seeds being dispersed effectively.

RV Lecture Slides

Aquilaria sinensis is a **canopy tree** in tropical forests. When its fruits mature:

The **capsules open high in the canopy**

Seeds are suddenly exposed to:

Intense solar radiation

High temperature

Low humidity

Under these conditions, seeds **lose viability very fast**

So unlike many fleshy fruits:

The plant **cannot afford slow dispersal**

Even hours of delay can be lethal

Why “normal” dispersers won’t work

Birds or mammals:

- Often require:

- Visual cues
- Large, nutritious pulp
- Time to locate and handle fruits

- May arrive **too late**

Wind dispersal:

- Unreliable in dense forest canopy

So *Aquilaria* needs:

Immediate, guaranteed removal of exposed seeds

In Aquilaria sinensis, rapid seed dispersal is achieved through emission of highly volatile short-chain compounds that attract Vespa hornets, enabling immediate seed removal from a lethal canopy environment rather than prolonged visual advertisement to frugivores.

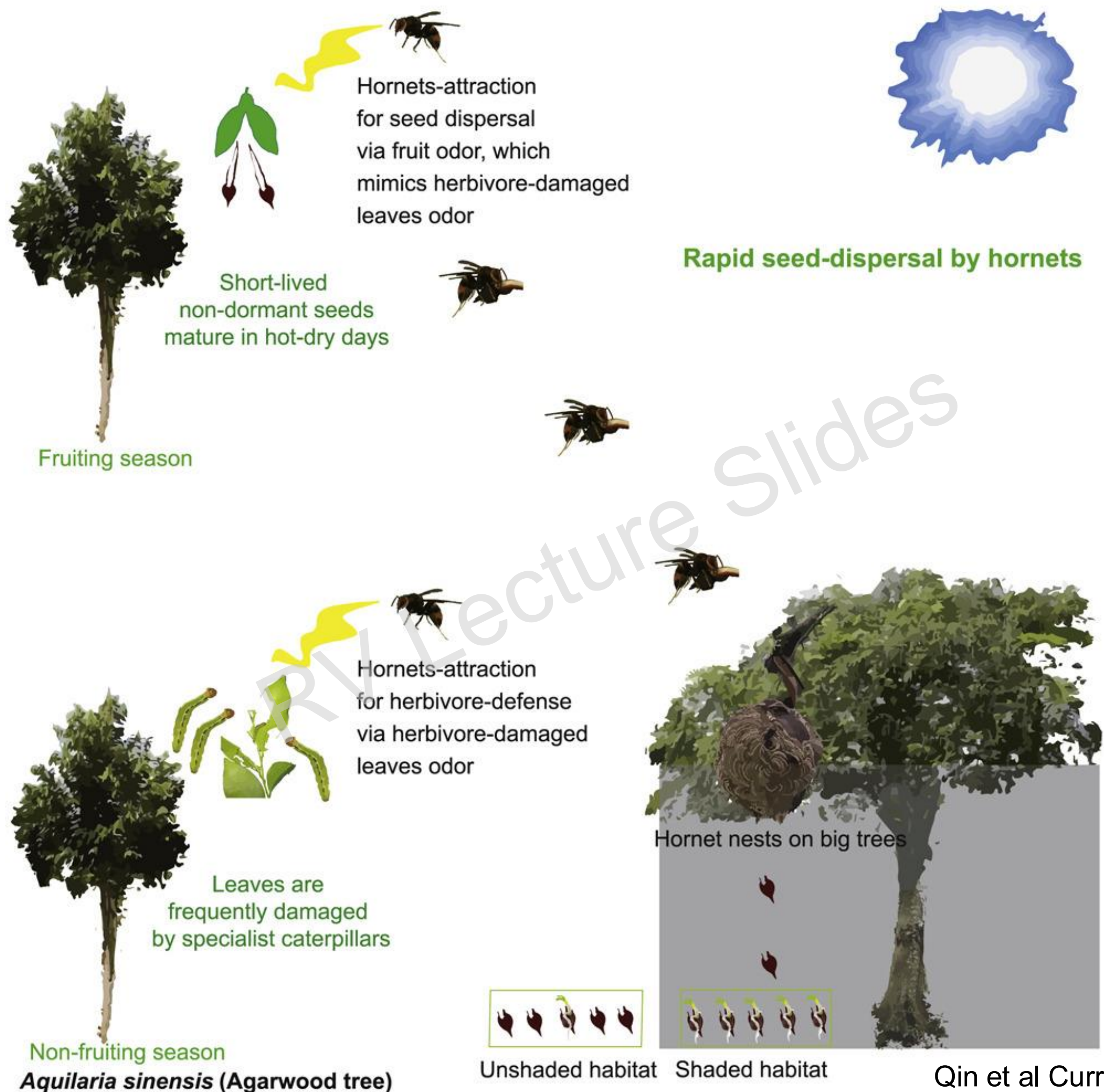


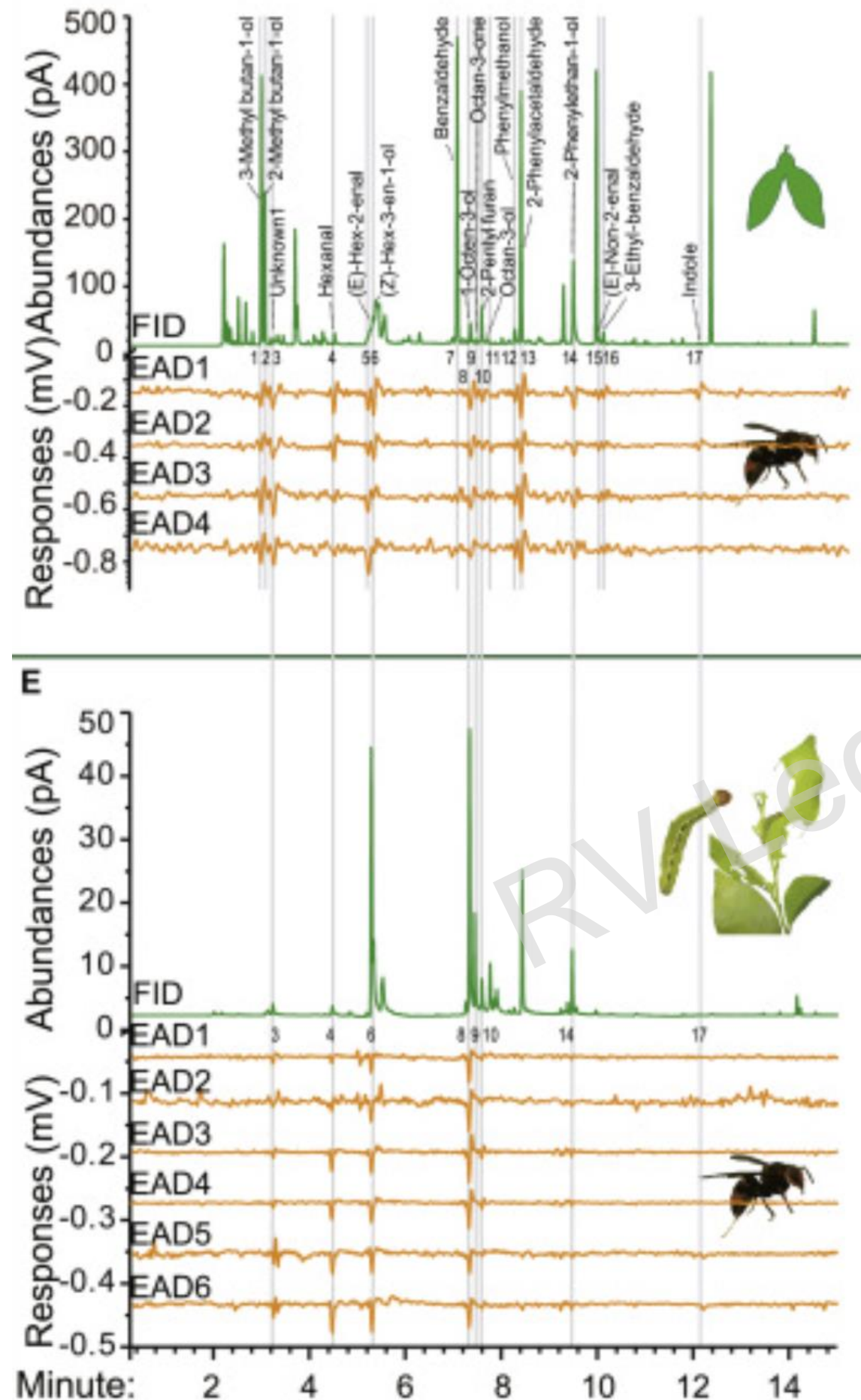
When ripe capsules of **Aquilaria sinensis** open in the canopy:

- Seeds become **fully exposed**
- The fruit releases **HIPV-like volatile odors**
- **Vespa** hornets arrive rapidly

The hornets then:

- **Grasp the seeds with their mandibles**
- **Fly off carrying them** away from the canopy
- Often drop them elsewhere or abandon them after inspection





The odor from fruit capsules mimics that of herbivore-damaged leaves of *A. sinensis* and results in the rapid attraction of hornets

This study demonstrates a need for extremely rapid seed dispersal in [*Aquilaria sinensis*](#), where the seeds are exposed to the hot, dry, and rapidly lethal canopy environment of a tropical forest. This need is fulfilled by rapid removal by *Vespa* hornets, attracted by highly volatile, short-chain carbon compounds produced by the ripe fruit capsules.

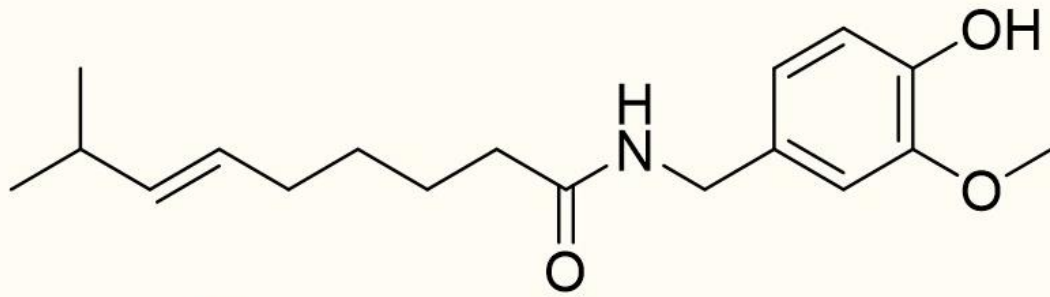
This system shows that:

- **Visual signals are not always optimal**
- **Olfactory signaling can dominate** when time is critical
- Seed dispersal traits evolve to solve *specific physical problems*, not just to attract animals

RV Lecture Slides



How can plants gain selectivity in seed dispersal?

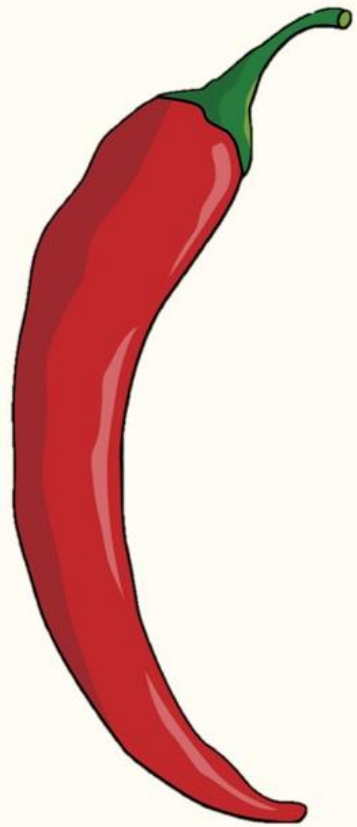


CAPSAICIN

The compound that, interacting with the TRPV1 receptor, is responsible for the pungency of hot chili peppers and related species and varieties

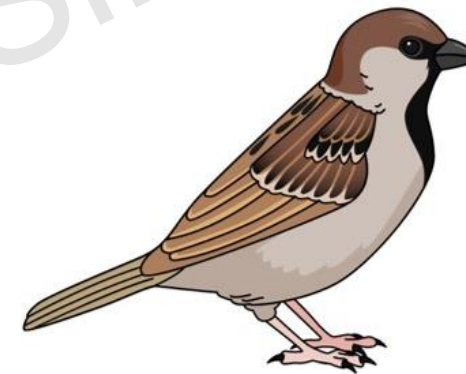
Produced by *Capsicum* species

- ✓ Found only in the fruits
- ✓ Abundant in the placental tissue
- ✓ Highest concentration in ripe fruits

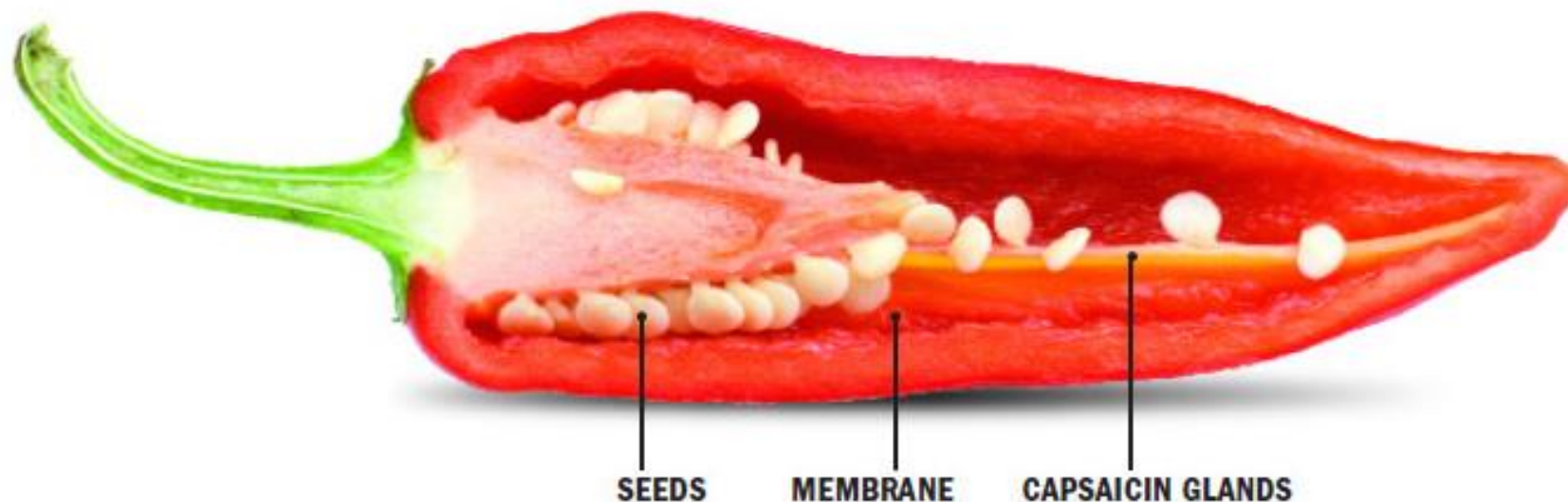


Functions in pepper fruits

- ✓ Protection from microbial infections and from the attack of several invertebrates
- ✓ Selectively discourages seed predators, while avoiding deterring beneficial dispersers (in accordance with the directed-deterrence hypothesis)



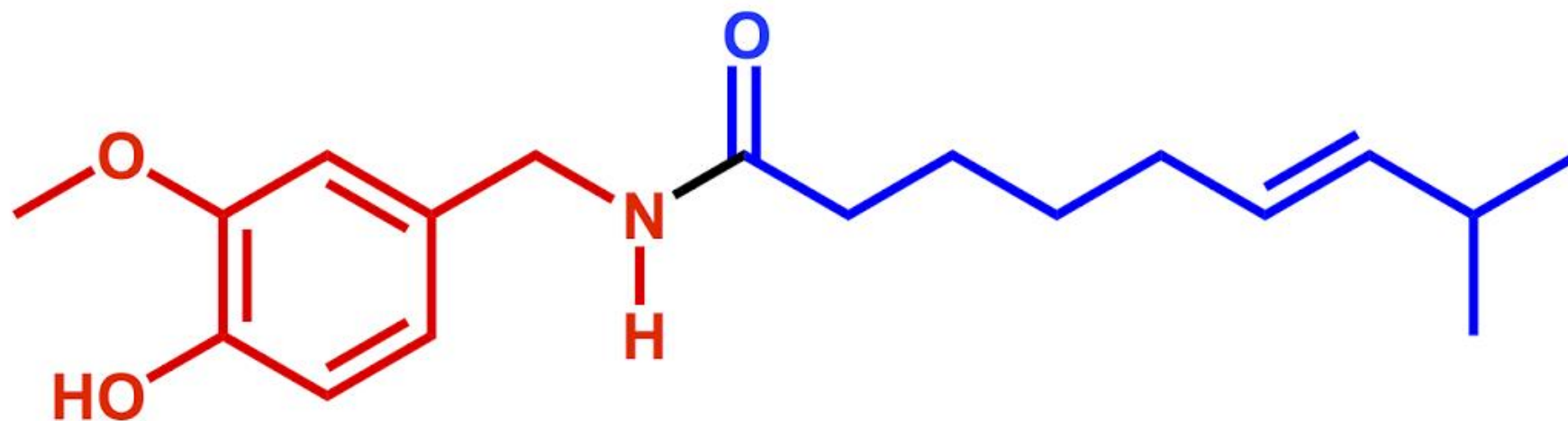
Images created with mindthegraph

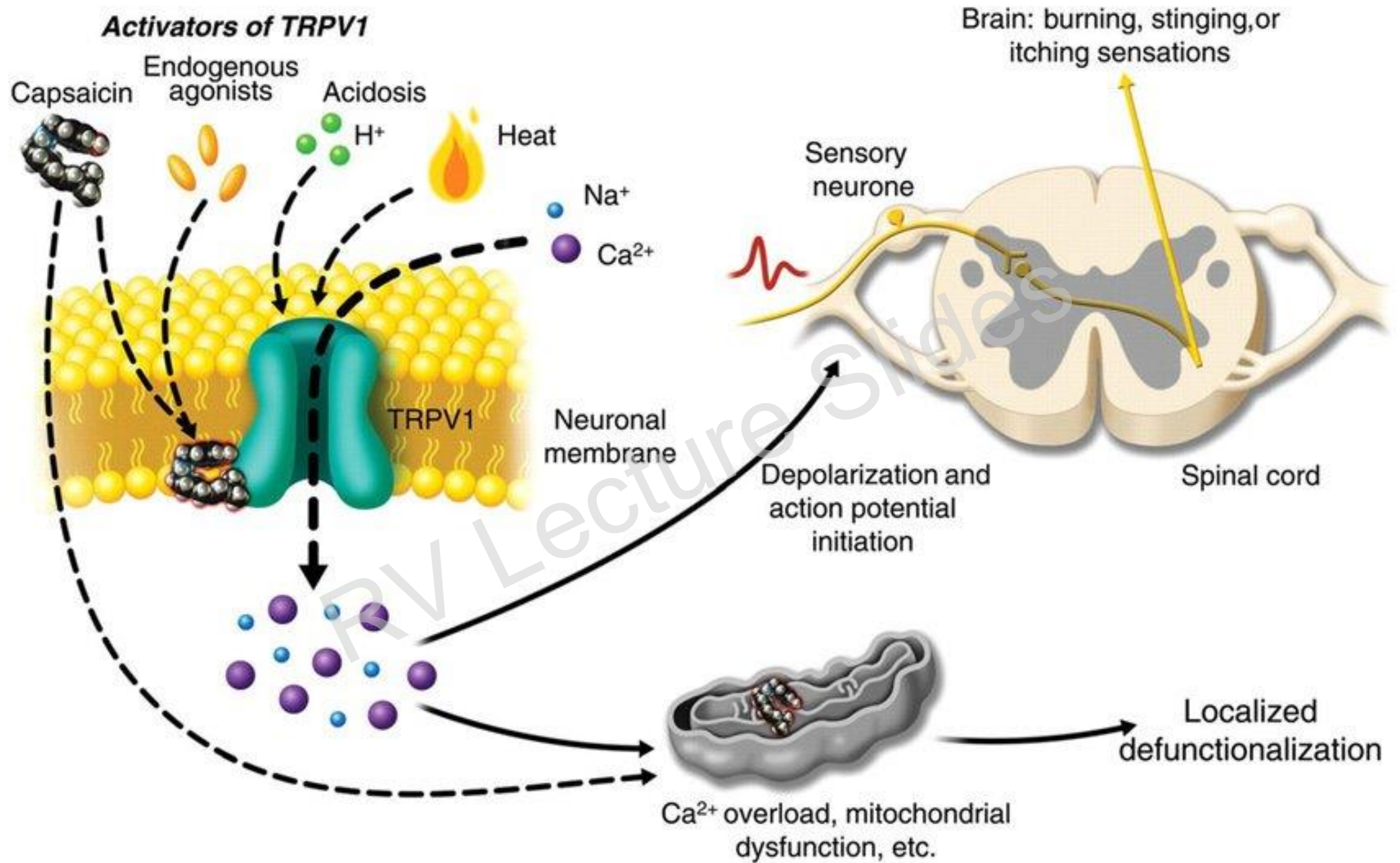


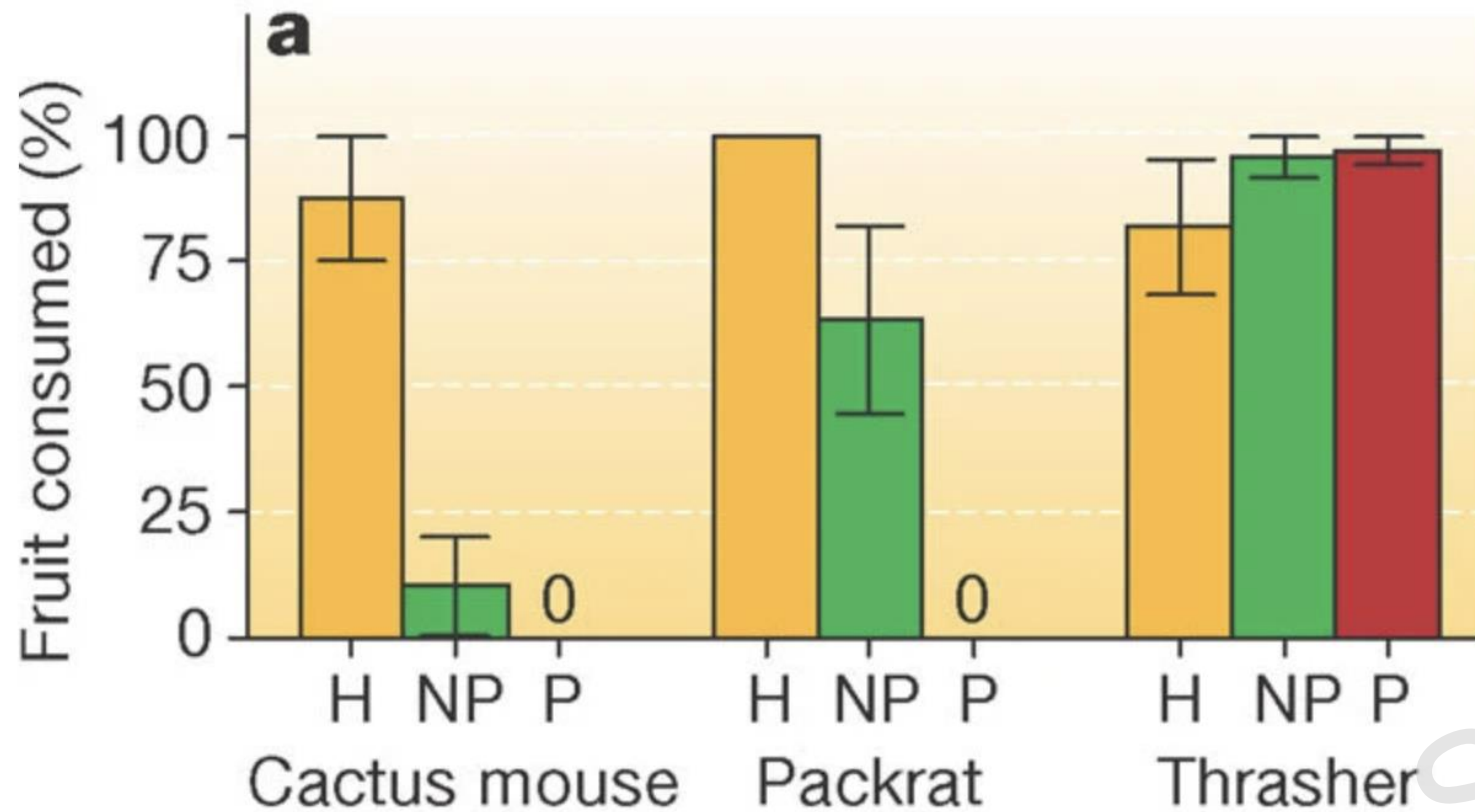


Vanilla-like part

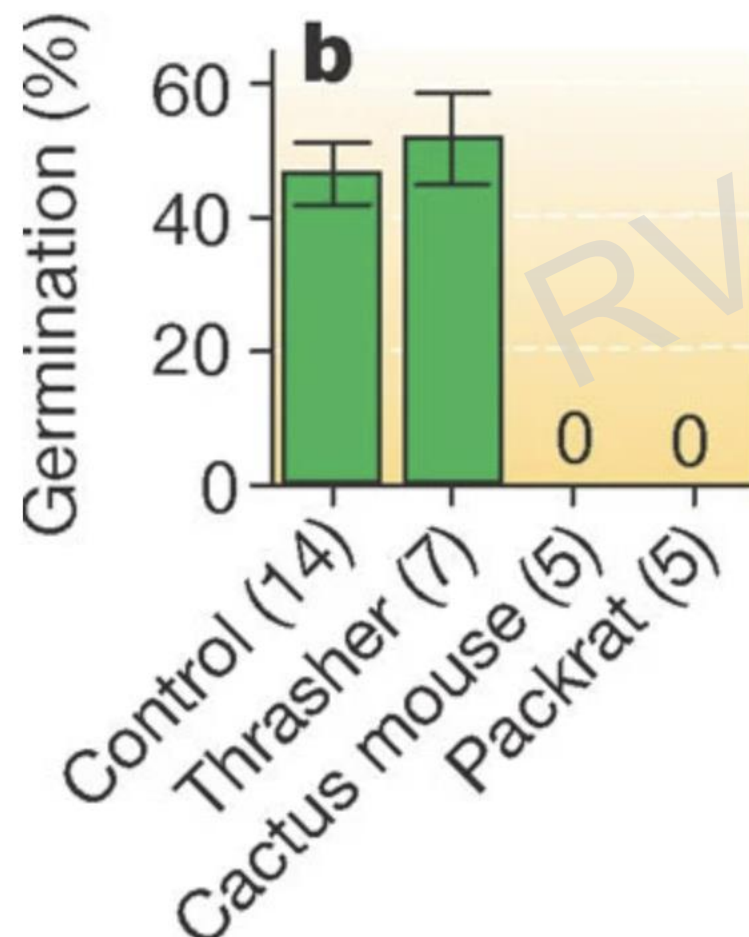
Fatty acid part







In captive feeding-preference trials, the two most common mammals readily consumed hackberry fruit (H); consumed non-pungent chillies (NP) only if they had not first sampled pungent chillies; and always avoided pungent chillies (P).



Seeds from non-pungent chilli fruit ingested by cactus mice or packrats did not germinate, primarily because both mammal species rarely passed whole seeds.

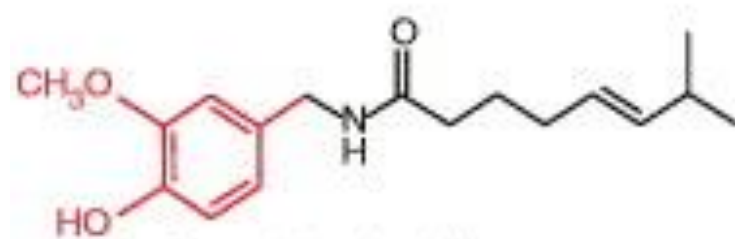
The combination of **chemical-mediated 'filtering'** of vertebrate seed predators, together with directed dispersal by thrashers, creates a pattern of plant recruitment that is beneficial to wild chilli plants: mammalian seed-predation is eliminated

Directed deterrence by capsaicin in chillies

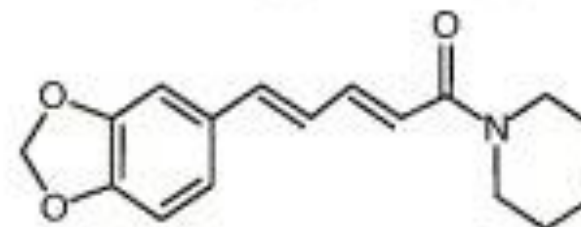
[Joshua J. Tewksbury](#) & [Gary P. Nabhan](#)

Nature **412**, 403–404 (2001) | [Cite this article](#)

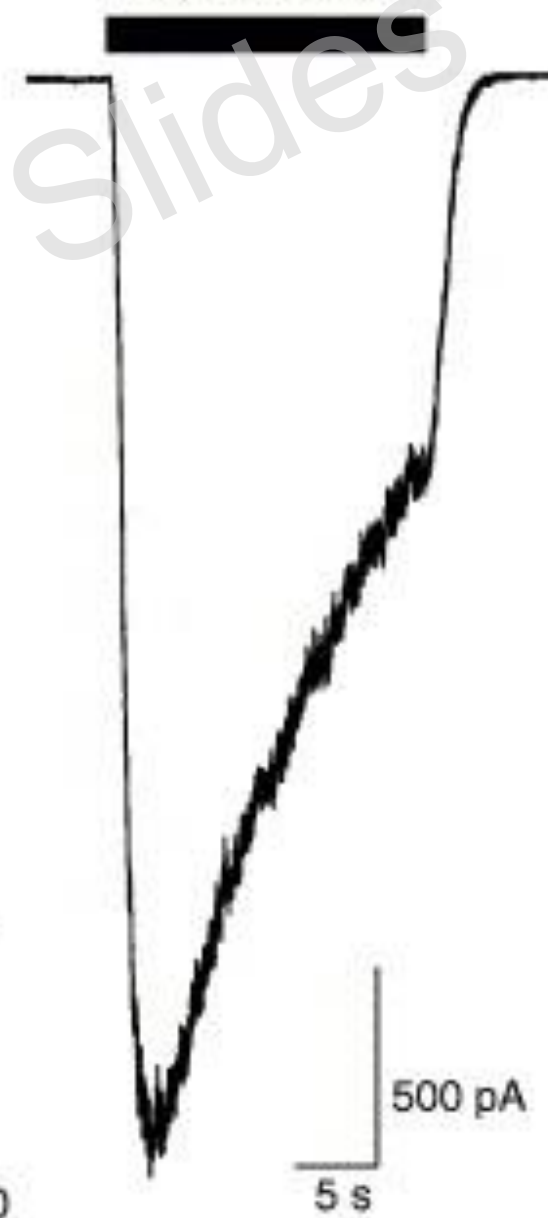
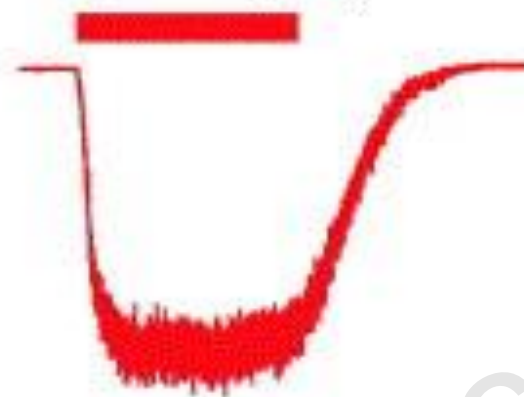
(a)



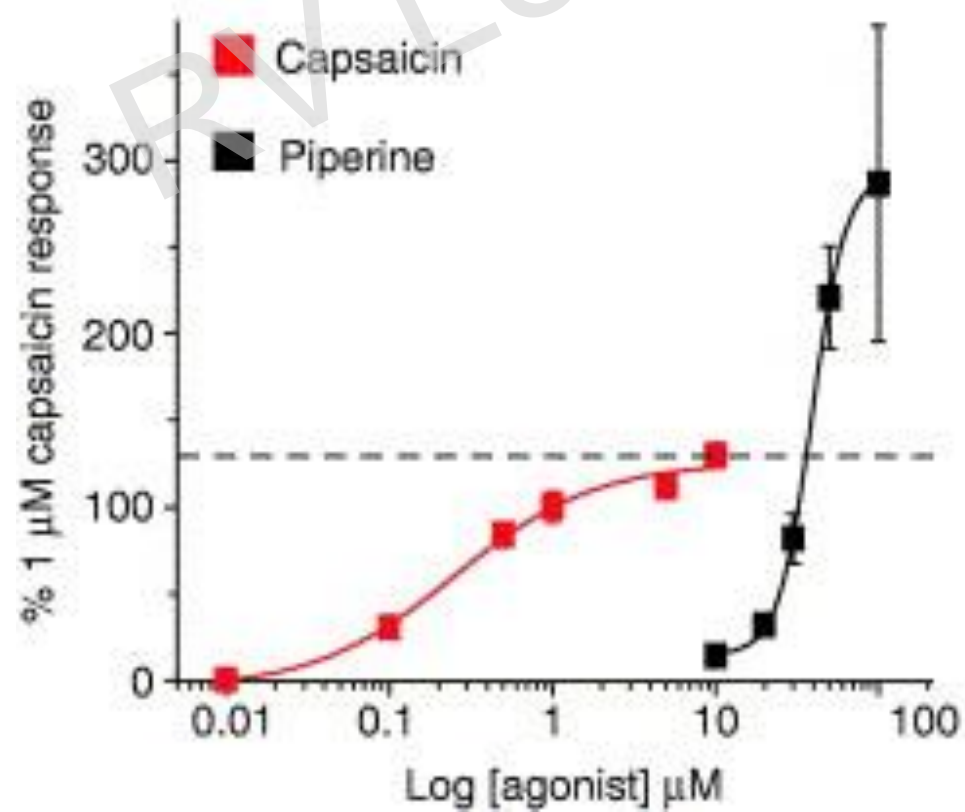
Capsaicin (1 μM)



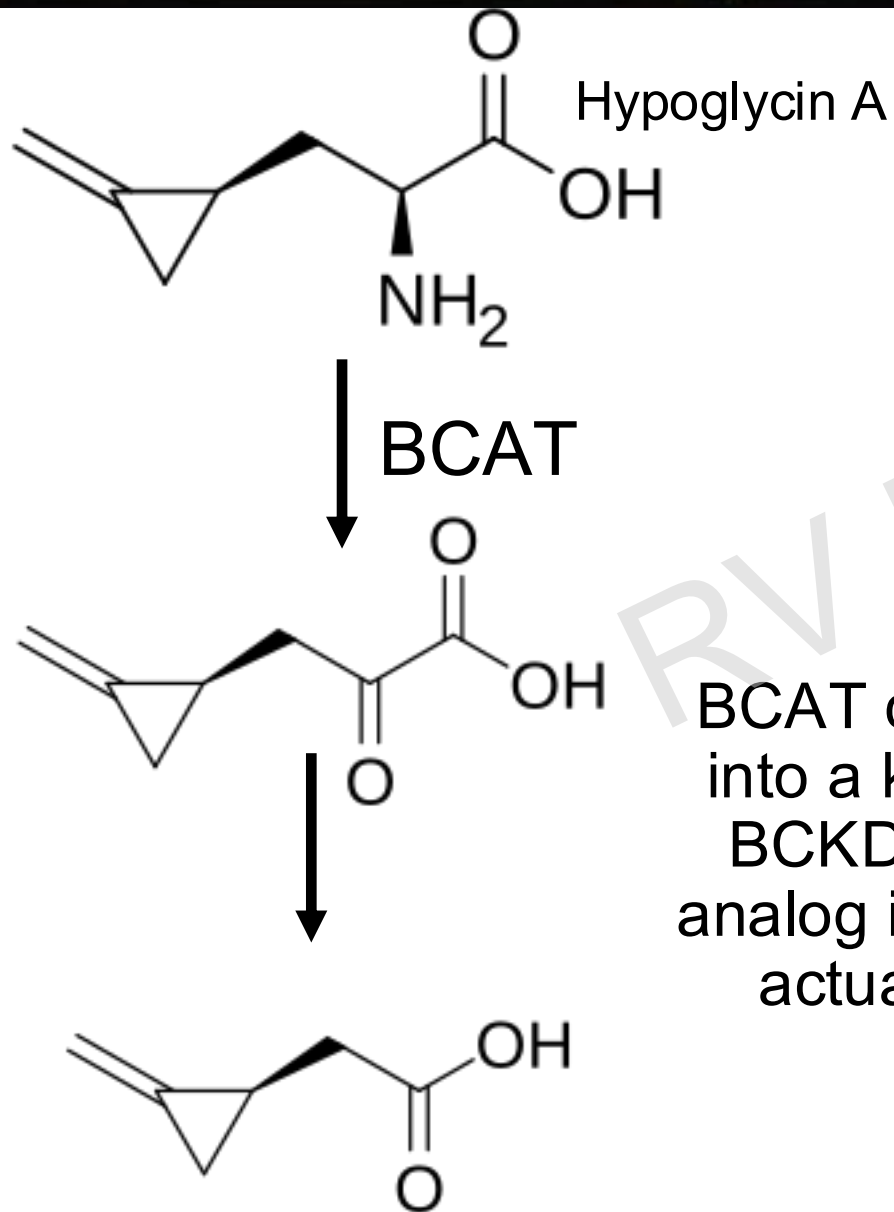
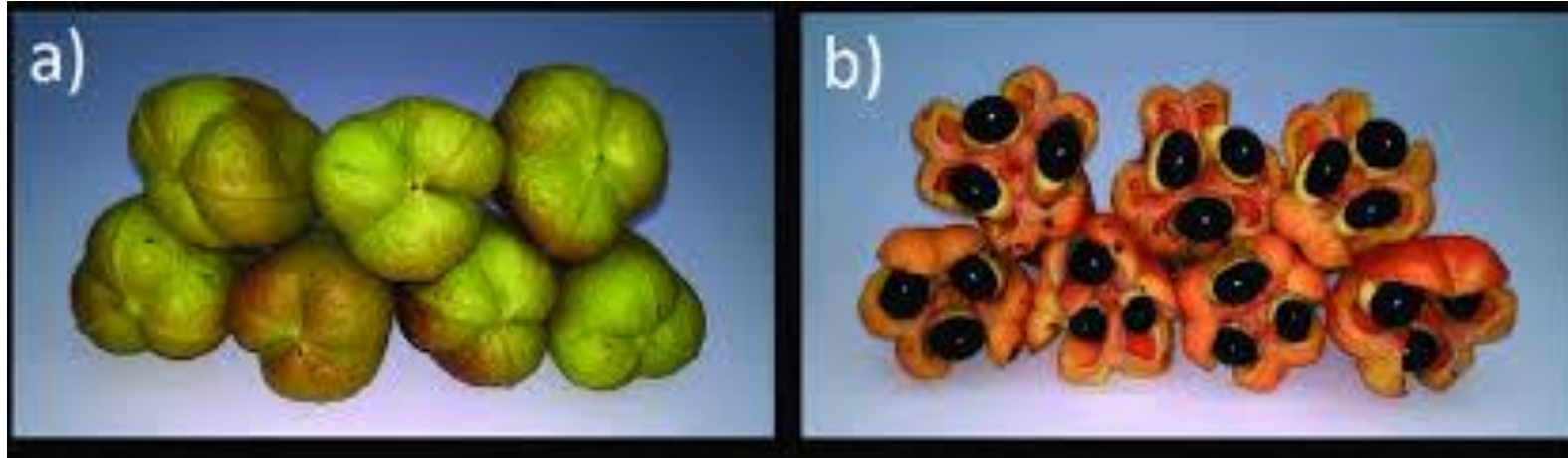
Piperine (100 μM)



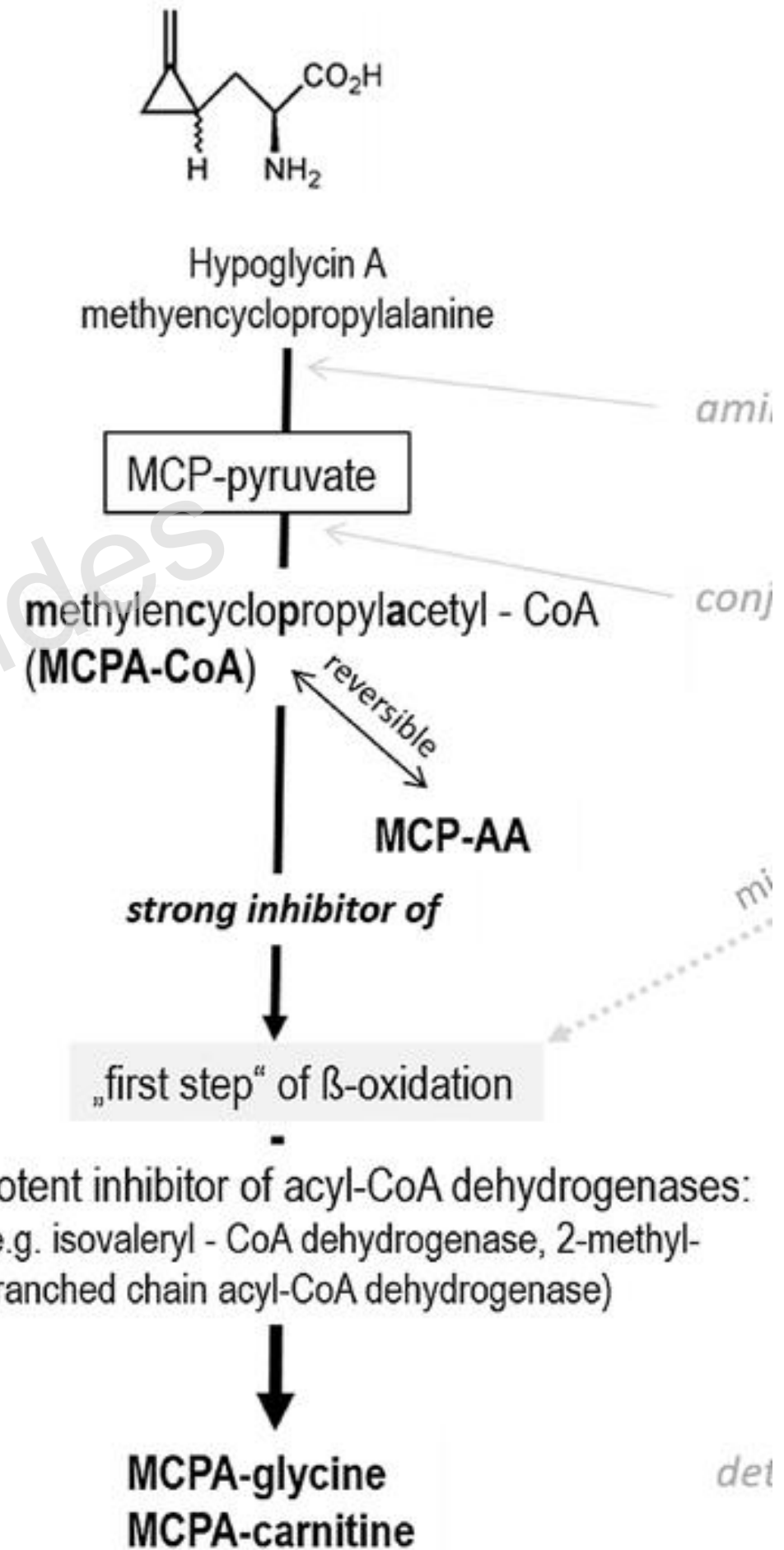
(b)

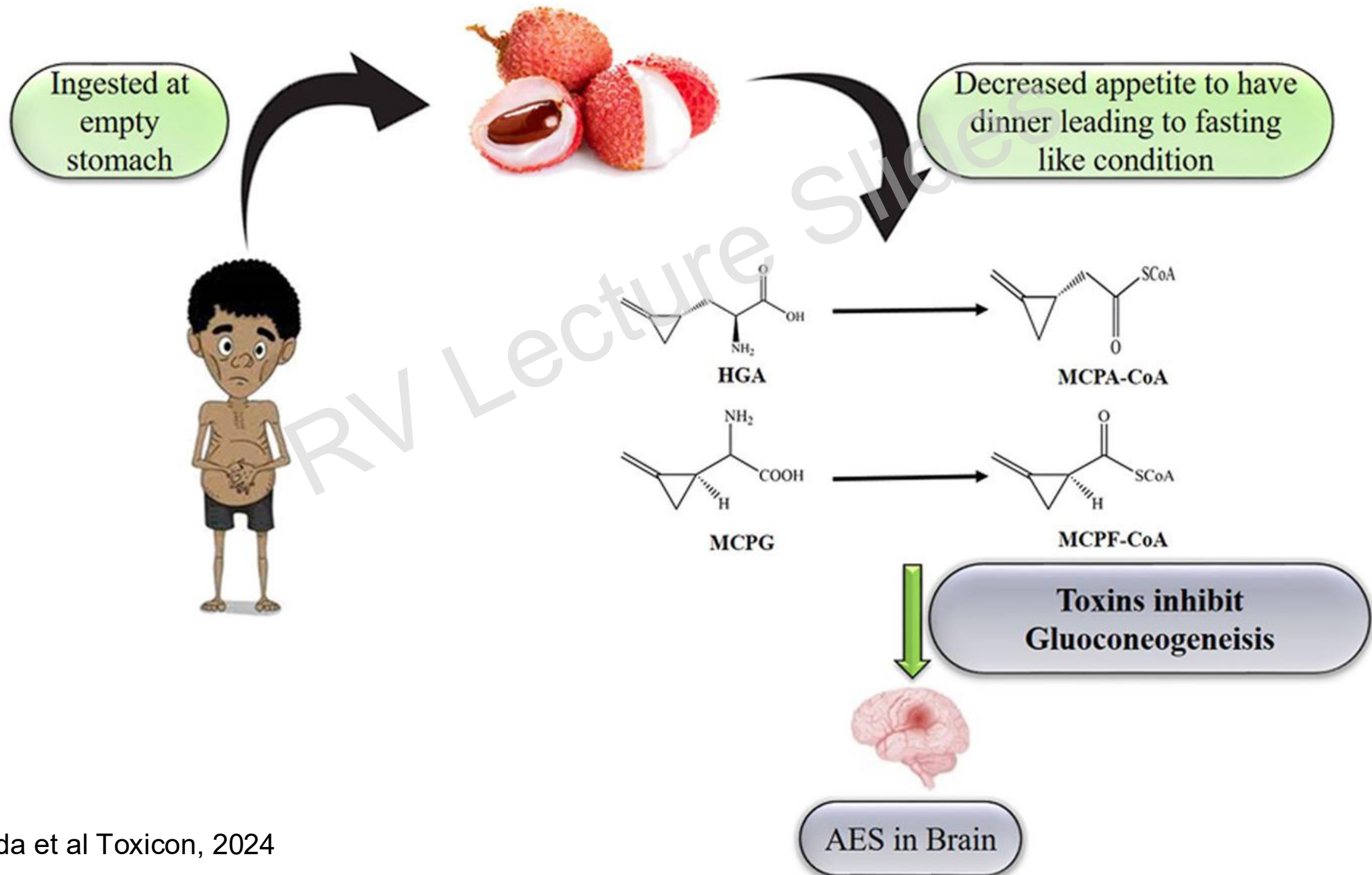


Blighia sapida

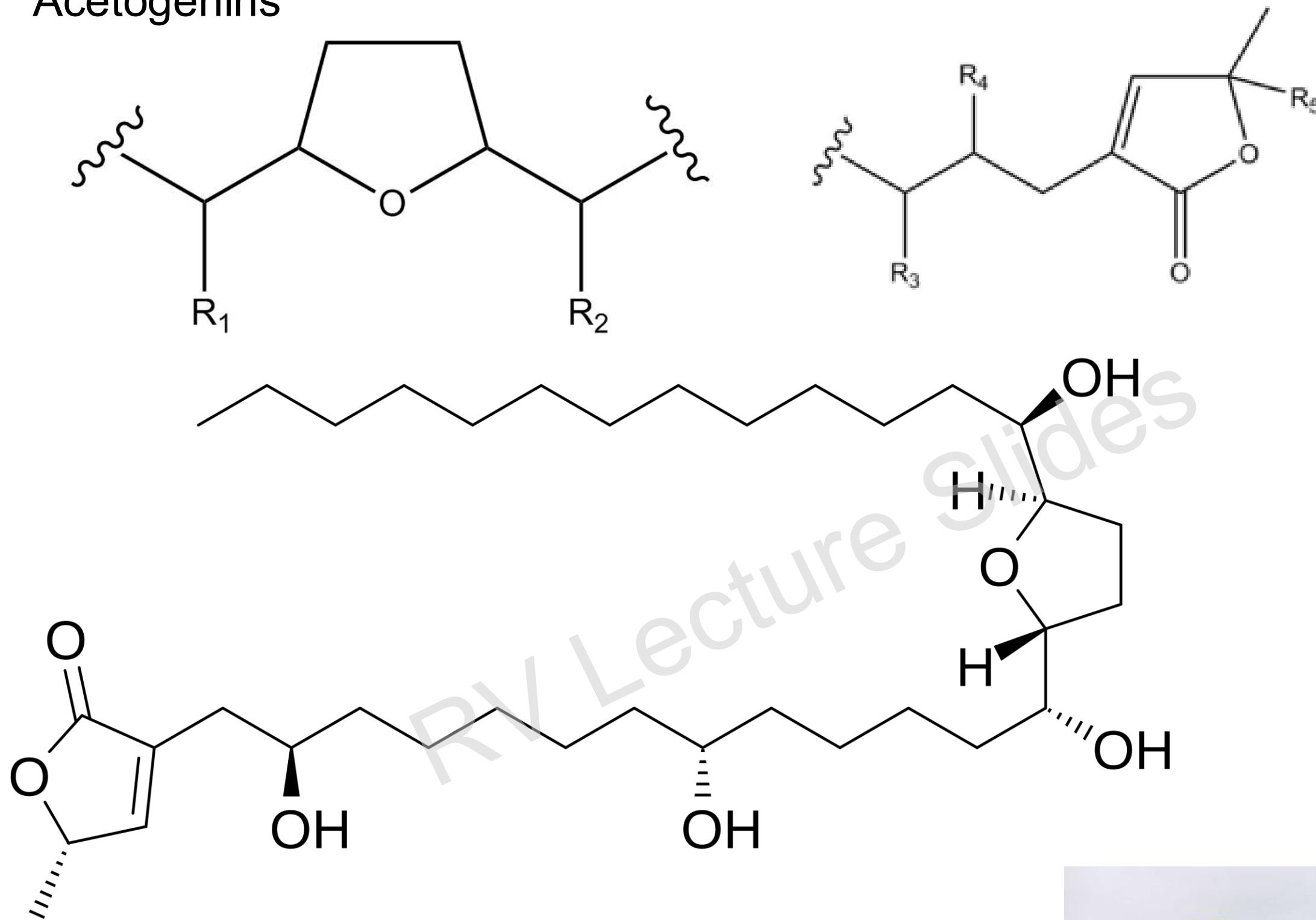


BCAT converts hypoglycin A into a keto analog, and *then* BCKDC converts that keto analog into MCPA-CoA — the actual metabolic poison.





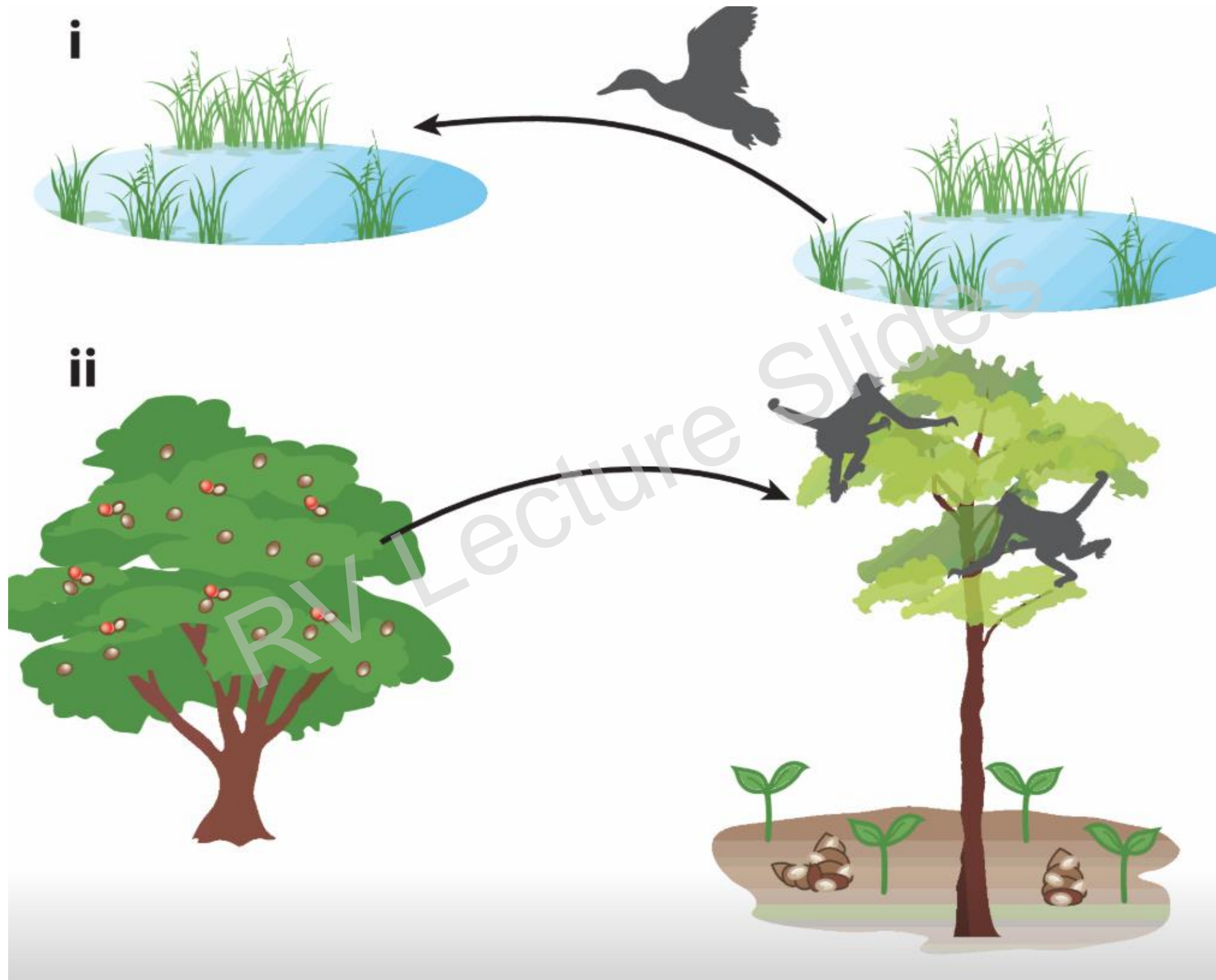
Acetogenins



ACG are known to block NADH ubiquinone reductase (Complex I) of the respiratory chain, directly affecting the electron transport in the mitochondria, causing ATP levels to decrease and cells to undergo apoptosis



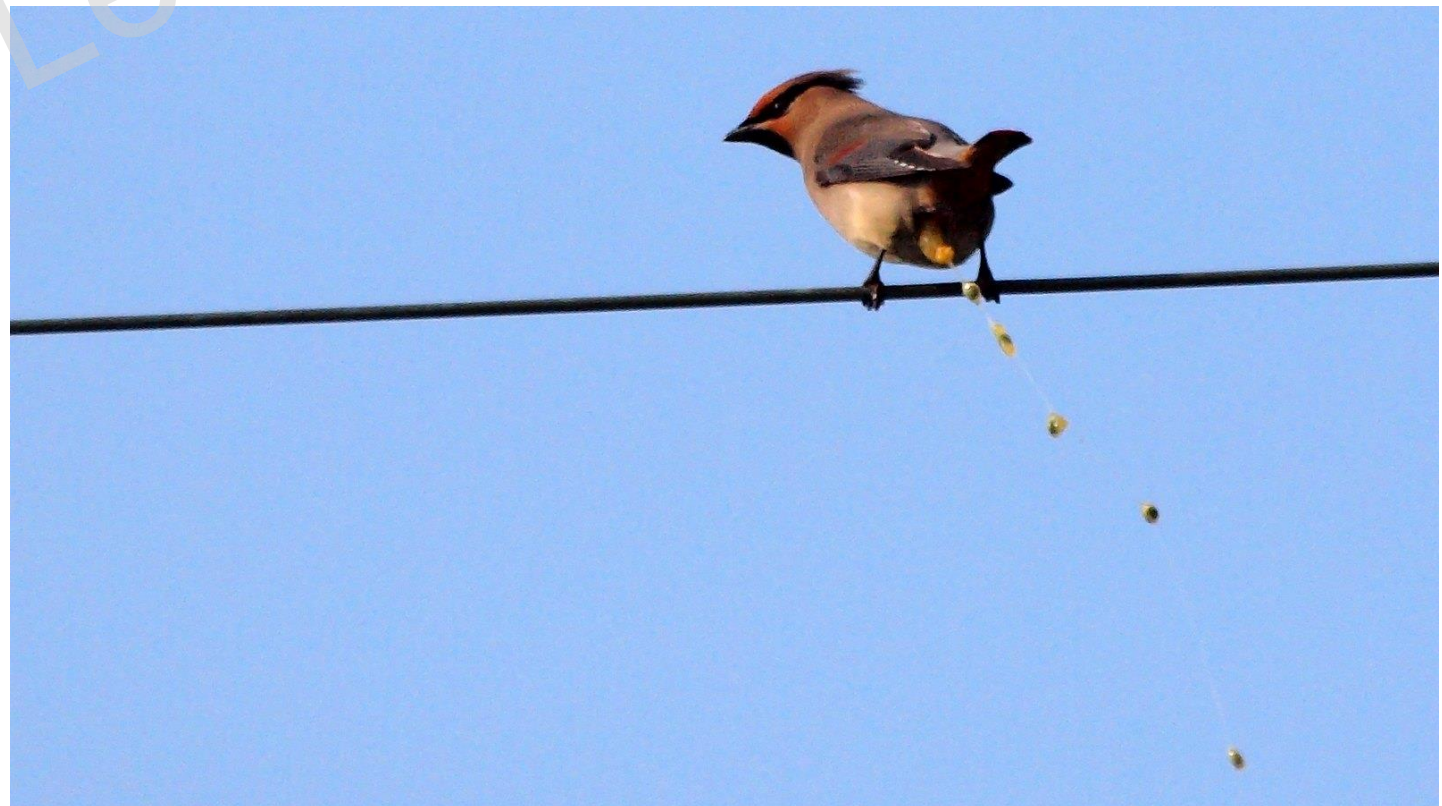
Can plants direct dispersal?



A recurrent theme in the study of seed dispersal mutualisms has concerned the degree to which plants can influence disperser behavior in order to maximize their own reproductive success.



In many bird species large seeds are voided by regurgitation while smaller ones are defecated. As a result, small seeds are retained longer in the gut and might travel farther from the parent plant, even though longer retention might reduce seed viability.



Why plants make laxatives when they don't get constipation?





***Witheringia solanacea* (Solanaceae), is a common pioneer in ranges from Mexico to Brazil.**

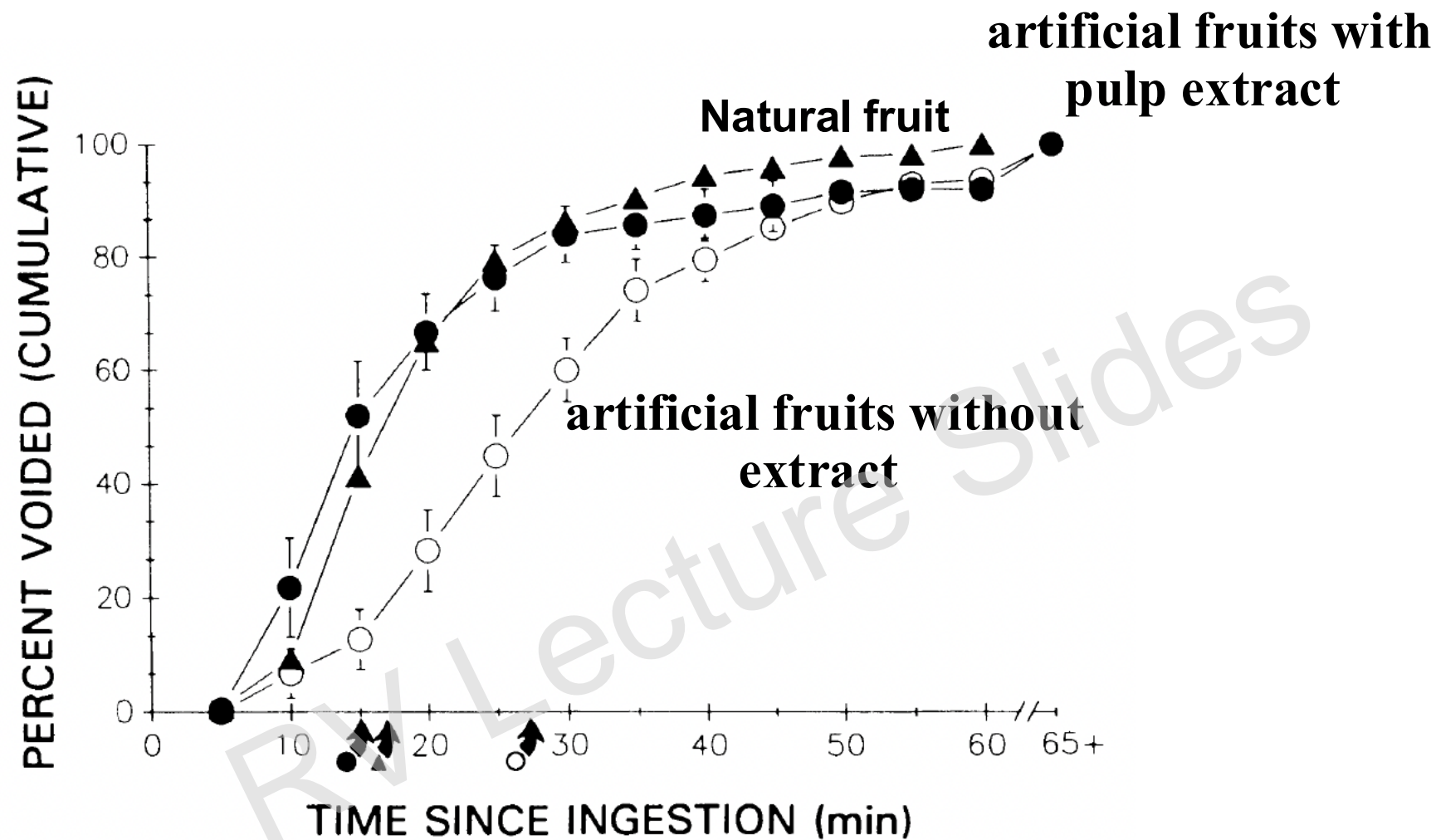
Black-faced Solitaire, *Mvadestes melanops* (Muscicapidae: Turdinae)



The authors constructed artificial "fruits" from a solution of 2.0 g agar, 7.5 g glucose, 7.5 g fructose, 100 mL boiling water, and 45 drops of red food coloring. The hot mixture was injected into a spherical mold (7.75 mm diameter) containing 10 *W. solanacea* seeds.

Half of the artificial fruits so produced were soaked in a small amount of crude extract from *W. solanacea* fruits (prepared by simply crushing natural fruits and discarding the seeds, pulp, and fruit skins) overnight before retention time experiments. Thus they obtained two types of artificial fruit identical in all respects save one: the presence/absence of soluble chemicals from *W. solanacea* pulp.

Birds consumed 1-8 (median 2.8; n = 48) fruits during the 5-min ingestion period, invariably swallowing them whole. In all cases, the first seeds emerged within 10 min of ingestion, and ~90% had emerged by 45 min.



Seed passage rates from artificial fruits with pulp extract were very similar to those from natural fruits. In contrast, seeds from artificial fruits without pulp extract were passed significantly more slowly (median passage time ~27 min) than those with extract (~15 min)

Germination success of *W. solanacea* seeds was strongly affected by gut passage. On the one hand, proportional germination was higher among seeds passed through Black-faced Solitaire guts (62.8%; all retention times combined) than among those removed from fruits by hand (51.7%).

On the other hand, among gut-treated seeds germination success decreased monotonically with increased time spent in the gut, even though all seeds, regardless of retention time, appeared intact.

Thus, short periods of treatment in Black-faced Solitaire guts are advantageous, but this advantage diminishes rapidly with longer retention times.

This is the first study to demonstrate the existence of a soluble fruit chemical (or chemicals) that mediates seed passage rate: seeds from artificial fruits that also included a crude extract from *W. solanacea* pulp were voided much more rapidly than seeds from fruits without the extract

THE CHEMISTRY OF PLUMS & PRUNES

PLUM AROMA & WAX BLOOM



BENZALDEHYDE



LINALOOL



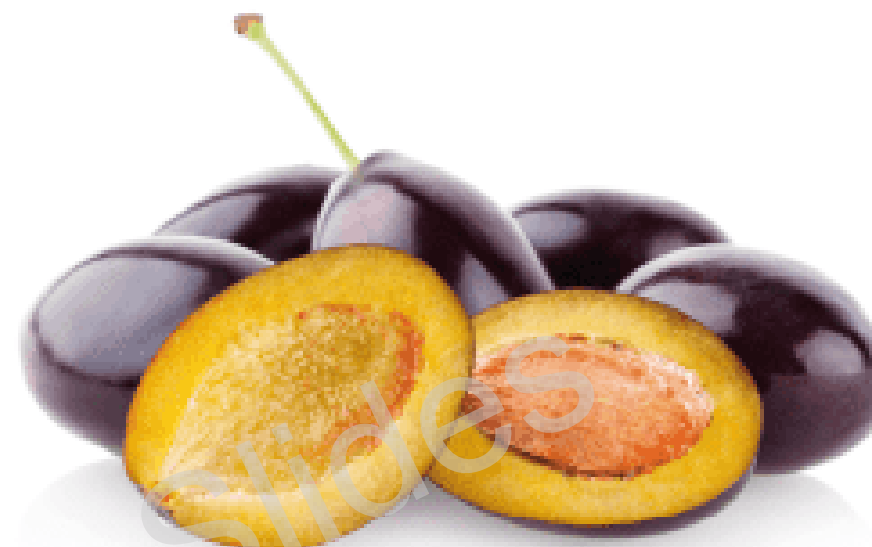
γ-DECALACTONE

The aroma of plums is down to a number of volatile compounds, which include benzaldehyde, linalool, ethyl nonoate, methyl cinnamate, & γ-decalactone. Several six-carbon alcohols, aldehydes, and esters also contribute.

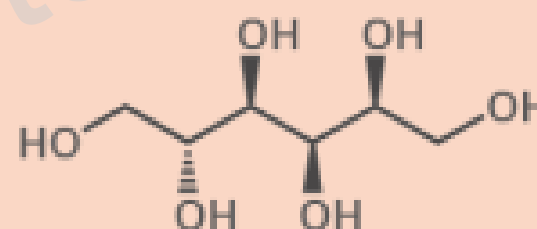
The dusty white coating visible on many plums is referred to as a 'wax bloom'. This bloom consists of long chain alkanes and alcohols (mainly those containing 29 carbon atoms), and adds to the flavour of the plum by trapping compounds such as nonanal.



NONANAL



WHY DO PRUNES HELP WITH CONSTIPATION?



SORBITOL

PRUNES

15g

CHEWING GUM

30g

VS

(SORBITOL CONTENT PER 100 GRAMS)

Prunes are dried plums, and are often cited as a home remedy for constipation. This is due to their relatively high natural levels of the known laxative compound sorbitol (approximately 15g per 100g). Sorbitol is also responsible for the laxative effect of some chewing gums. Phenolic compounds, such as neochlorogenic acids, and the high fibre content of prunes may also aid the laxative effect.





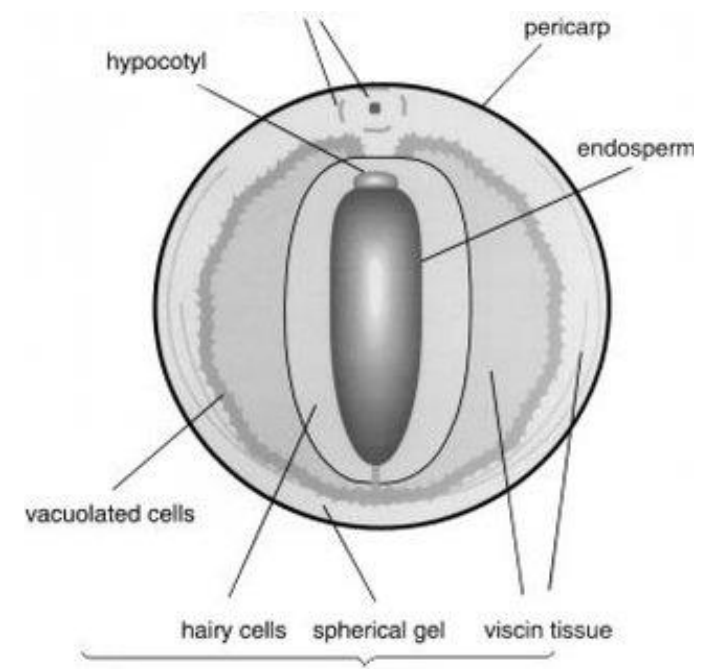
Merry Christmas



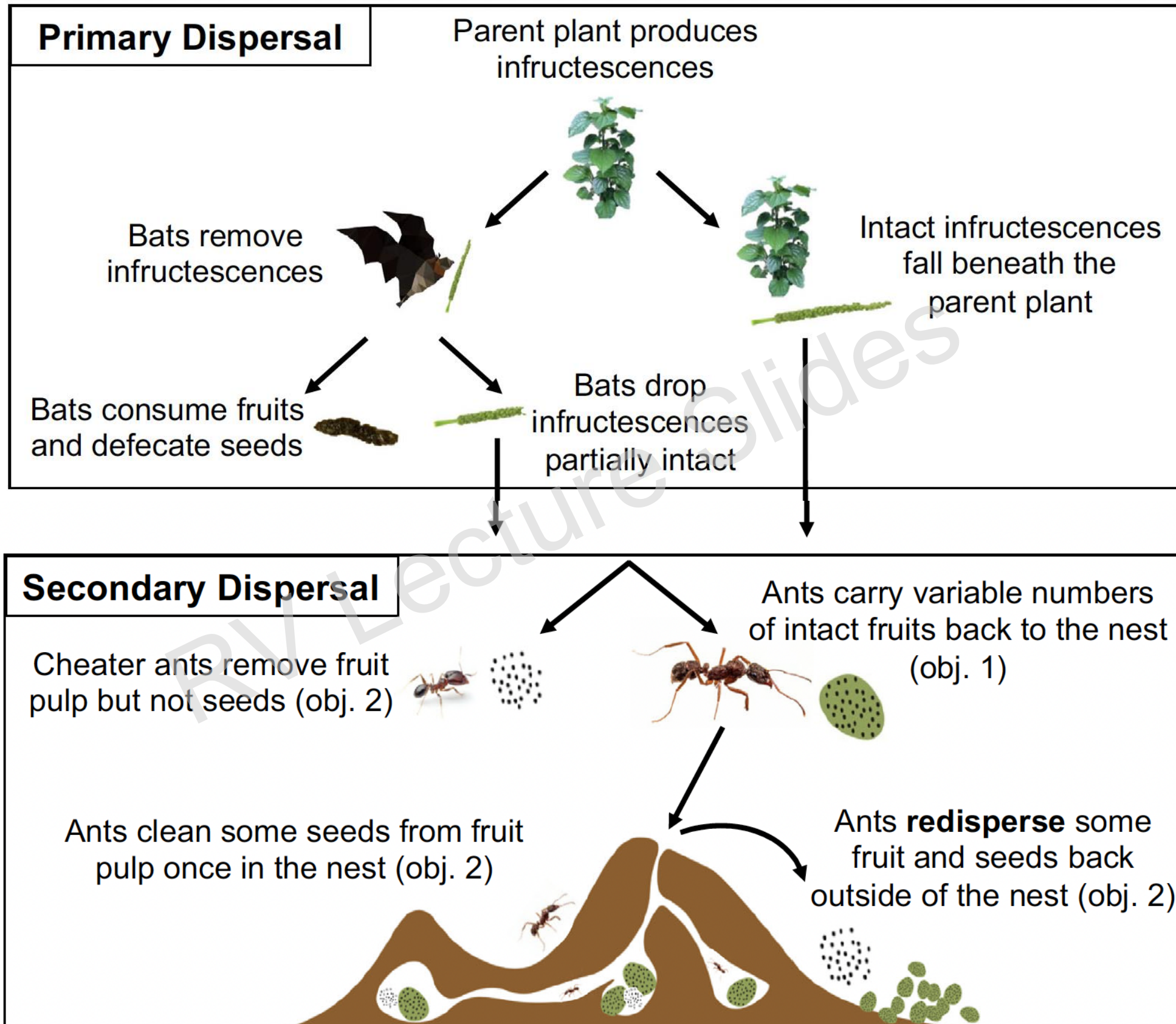


Each mistletoe berry can produce up to two metres of a gluey thread called viscin. It allows the seeds of this parasitic plant to stick to and infect host plants.

Approximately 90 bird species from 10 families are considered mistletoe fruit specialists. The sticky viscin helps the propagules to stick to the branches.

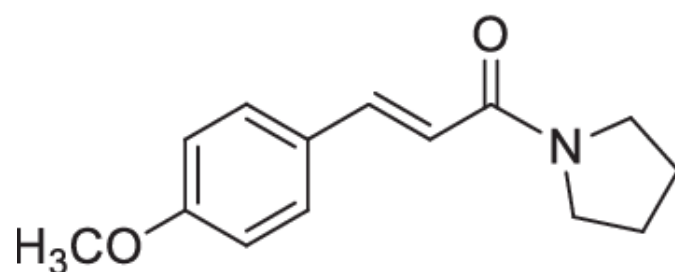


Piper, bats, and ants

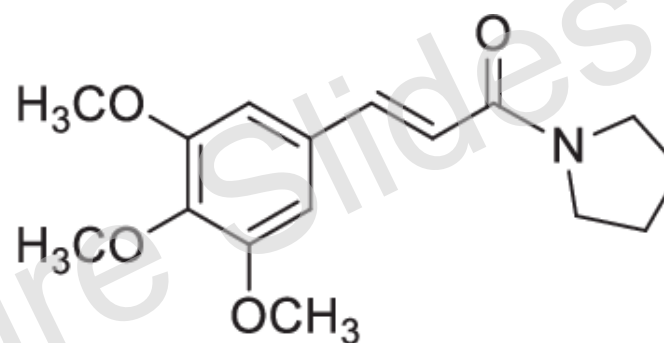


Piper fruits are rich in **Piper amides** (e.g. piperine-like molecules):

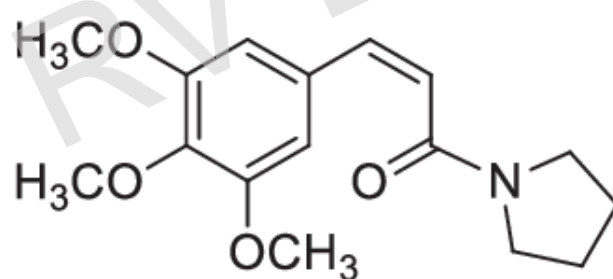
- Bitter
- Neuroactive
- Deterrent to many insects and vertebrates
- Structurally similar to TRP-channel-active compounds



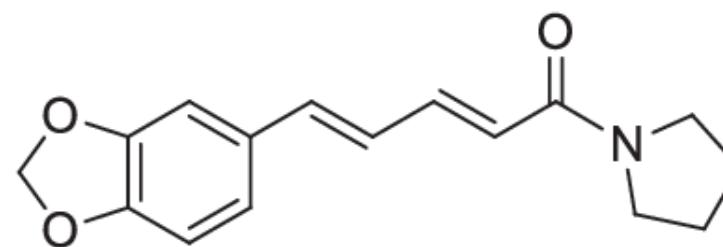
Piperlotine A



Piperlotine C



Piperlotine D



Piperiline

When amide concentrations are high:

- Bats **visit less frequently**
- Feeding bouts are **shorter**
- Bats chew briefly, then **spit out fruit pulp**
- Seeds are dropped closer to feeding roosts

When amide levels are lower:

- Longer feeding
- More seeds swallowed
- Greater gut passage → longer dispersal distances

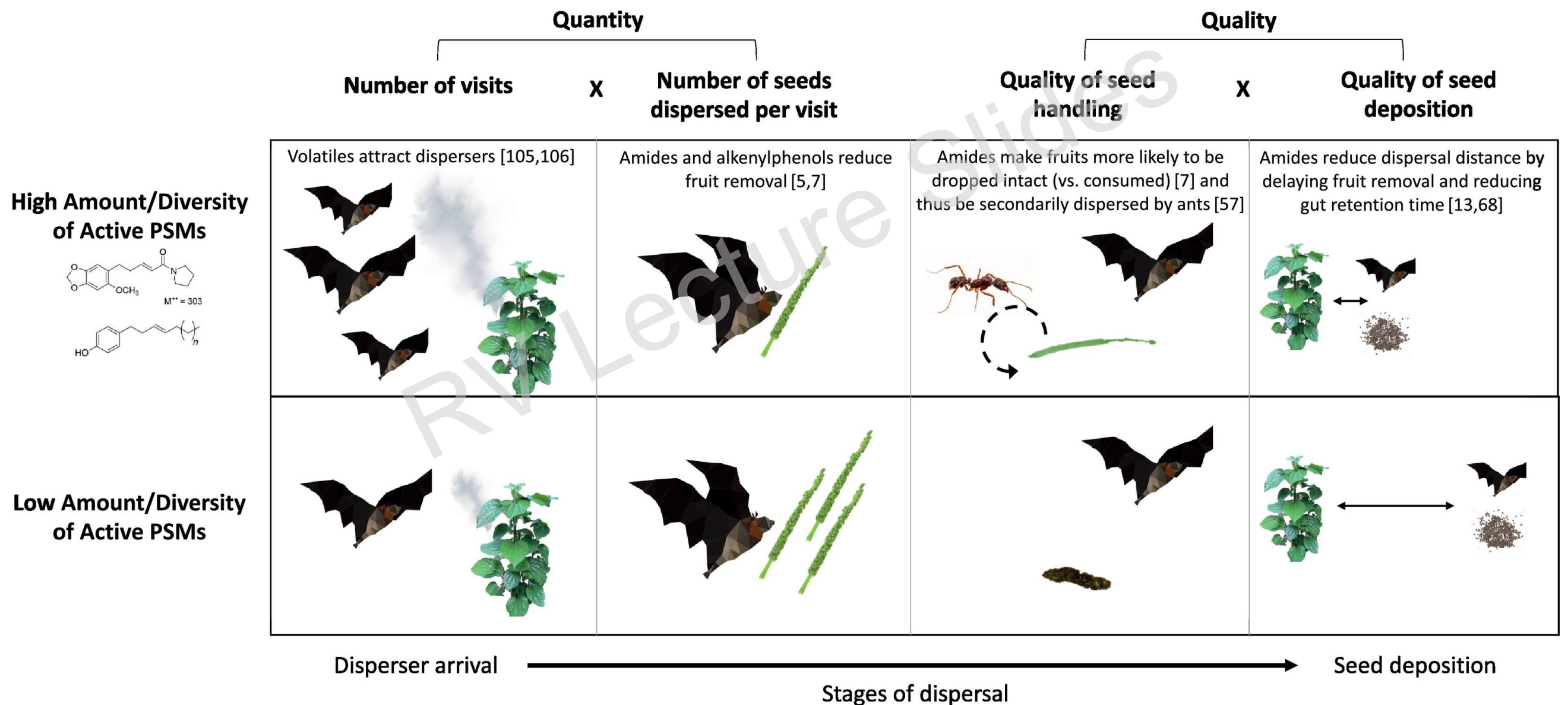
So chemistry tunes:

- **Dispersal quantity** (how many seeds moved)
- **Dispersal distance**
- **Seed deposition pattern**



Plant chemicals shape seed dispersal: disperser arrival as well as seed deposition.

In the Piper–bat–ant system, fruit amides deliberately induce bats to drop seeds, creating resource patches that recruit ants as secondary dispersers and ultimately increase seed survival and dispersal quality — turning an apparent loss into a fitness gain.



Ants as secondary seed dispersers



Elaiosome



Leptogenys processionalis



Solenopsis geminata



Monomorium indicum



Camponotus compressus



Aphaenogaster beccarii



Pheidole grayi

Ants are attracted to seed volatiles and remove elaiosome based on their fatty acid content

